



Simulation of rerouting phenomena in Dynamic Traffic Assignment with the Information Comply Model



Rafał Kucharski^{a,*}, Guido Gentile^b

^a Department of Transportation Systems, Cracow University of Technology, ul. Warszawska 24, 31-155 Kraków, Poland

^b Dipartimento di Ingegneria Civile Edile e Ambientale – DICEA, Università degli Studi di Roma “La Sapienza”, via Eudossiana 18, 00184 Rome, Italy

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ABSTRACT

We present the Information Comply Model (ICM) which extends the framework for macroscopic within-day DTA proposed by Gentile (2016) to represent the rerouting of drivers wrt a single traffic event. Rerouting is reproduced as a two-stage process: first, drivers become aware about the event and its consequences on traffic; second, drivers may decide to change path. At each arc, unaware drivers have a probability of being informed by multiple ATIS sources (radio, VMS, mobile apps), which depends not only on device penetration rates, but also on users space and time coordinates wrt the position and interval of the event. At each node, aware drivers, who are somehow reluctant to change, may finally modify their path based on a random rerouting utility, which is composed of expected gains and avoided losses. ICM is thus capable of representing the evolution of rerouting phenomena in time and space when the information about a traffic event and its consequences on congestion spread among drivers and onto the network.

This way, ICM extends the concept of dynamic user equilibrium to a case of imperfect information related to availability and awareness rather than to individual perception, as well as to a case of bounded rationality with prudent drivers. Besides the model architecture and specification, this paper provides a workable methodology which can be applied both off-line for transport planning and in real-time for traffic management on large size networks.

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1. Introduction

Traffic events (vehicle accidents, road closures, demand peaks, etc.) happen on a daily basis on road networks. Most typical effects are related to local capacity reductions, or global flow increases, which may both cause oversaturation, congestion and then delays. Rerouting takes place when drivers, while travelling on the network, realize that the situation is atypical, and then decide en-trip to change their path, so as to partially avoid the negative consequences of the event.

Forecasting how events affect the traffic pattern is of particular importance, since proper monitoring (to assess impacts) and management (to take actions) rely on predictions. Due to events, both travel times and traffic flows change with respect to typical values, not only for their direct effects, but also as a consequence of their indirect effects. More specifically, the event impacts may propagate on the network both upstream because of queue spillback and sideways because of rerouting, causing further delays (Li et al., 2015). Modelling these effects on the supply side is relatively straightforward; if the route

* Corresponding author.

E-mail addresses: rkucharski@pk.edu.pl (R. Kucharski), guido.gentile@uniroma1.it (G. Gentile).

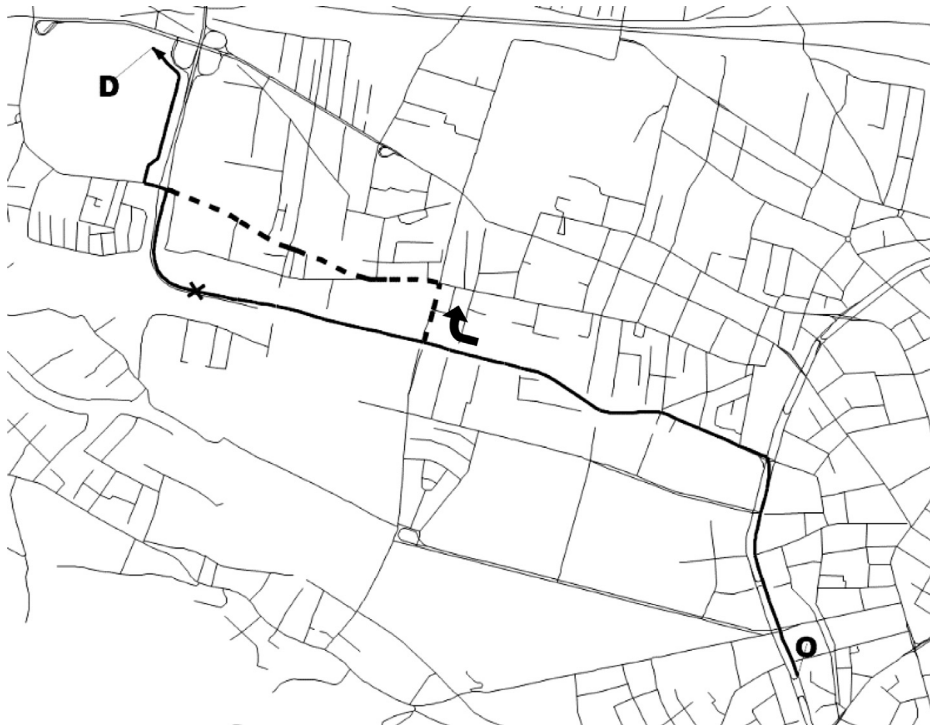


Fig. 1. Due to the event (marked with an X) driver reroutes en-trip from the typical (bold) path to the actually optimal path (dashed) at a decision point (marked with arrow).

choice pattern is assumed fixed, this can be accomplished through a dynamic network loading model (Corthout et al., 2011). It is instead much more challenging to represent the effects on the demand side, namely how drivers are first informed and can then react to unexpected events by changing en-trip their pre-trip route choice.

Some events are objectively unpredictable, and some other are only subjectively such. In both cases, not all drivers are informed in due time, i.e. before starting their trip. Moreover, drivers tend to be reluctant in changing their habitual path, which they clearly know better than currently optimal alternatives, so that modest delays are typically accepted without rerouting. Deficient information and prudential behaviour contradict the main assumptions of deterministic user equilibrium, i.e. perfect information and rational behaviour. Moreover, stochastic assignment is not of great help in reproducing the situation under analysis, because events produce perturbations of equilibrium which have an anisotropic gradient in space and time. Although several rerouting models were proposed in the literature, none of them allows at the same time to represent drivers' behaviour realistically and to simulate efficiently how traffic flows and travel times evolve due to the event on large networks.

Fig. 1 illustrates the *rerouting* phenomenon in an urban context. The path chosen pre-trip by the driver, that is optimal in terms of its *typical* costs, is not optimal anymore due to the unexpected event and to the delay that it caused. However, the driver may decide to change his/her path, to one that is optimal in terms of *actual* costs, only after receiving information about the traffic event and then realizing that he/she can partially avoid its negative consequences.

Other authors call this phenomenon *adaptation* (Gao et al., 2010), *recourse* (Unnikrishnan and Waller, 2009) or *en-route switching decisions* (Hu and Mahmassani, 1997); if path changing is limited to pre-trip choices, it is usually referred to as route-swapping and handled in the context of day-to-day models (Cantarella and Watling, 2016). The relevance of rerouting has raised in the last years as the number of informed drivers has substantially increased with the growing market penetration of Advanced Traveller Information Systems (ATIS).

In this paper, we present the Information Comply Model (ICM), which reproduces rerouting as a two-stage process, where drivers, first become aware of the event with a certain probability at each arc, thanks to a given ATIS configuration of available information sources (radio, VMS, internet apps, etc.), and second decide whether to change to an optimal path with a certain probability at each node, based on expected gains and avoided losses. Finally, rerouting drivers use their actual prediction of travel times and costs to choose their new path to destination. This schema results in a particular fixed-point formulation of Dynamic Traffic Assignment (DTA) where rerouting decisions are iteratively adjusted to reach an equilibrium which incorporates the above two-stage process.

We apply the ICM in the macroscopic DTA framework of Gentile (2016), which is here properly extended to represent rerouting. The resulting model has also been implemented as an add-in of PTV Optima, a commercial software solution

for traffic management (PTV SISTeMA, 2016); this allowed to test the proposed model on both illustrative examples and real-size networks, thus demonstrating its practical applicability and the soundness of results.

The paper is organized as follows. This introduction is complemented by a literature review and an outline of main contributions. In Section 2, the macroscopic DTA is introduced and its limitations to reproduce rerouting are identified. In Section 3, the Information Comply Model is presented, including the awareness model and the rerouting model. In Section 4, the ICM is casted into the proposed DTA framework. In Section 5, some numerical results illustrate the capabilities of the resulting model to represent rerouting also on real-size networks. Finally, conclusions are drawn and further research directions are outlined.

1.1. Background

Thanks to recent developments, DTA is now applied not only in off-line transportation planning, but also in real-time traffic management (e.g. PTV Optima, Gentile and Meschini, 2011; DynaMIT 2.0, Lu et al., 2015; AIMSUN online, Frisoet al., 2013). This, however, raises a number of issues, one of which is addressed in this paper: as Peeta and Mahmassani (1995) and later Dobler et al. (2012) pointed out, the usual equilibrium models, even if based on dynamic assignment methods, fail to reproduce atypical and unexpected situations where drivers reroute trying to avoid a traffic event.

The need for improvement is motivated by the fact that the state-of-the-practice presents two solutions to handle unexpected events within DTA, both of them intrinsically wrong. The first one is to solve a Dynamic User Equilibrium problem, assuming that the event and its consequences on travel times and traffic flows are perfectly known to drivers, which results in a predictive behaviour where users reroute before the event took place. The second one is to solve a Dynamic Network Loading problem, assuming that drivers will stick to their habitual path without rerouting, which results in the propagation of the demand flows on the network where the impacts of the event are taken into account only on the supply side. This is why an intermediate, more realistic model is looked-for and proposed in this paper.

The rerouting problem can be derived from the route choice problem (for a broad review of route choice models we refer to Prato, 2009). Initial formulations, where the arc costs are assumed fixed and deterministic, have been gradually improved and became applicable also to the case of en-trip rerouting. Polychronopoulos and Tsitsiklis (1993) address the shortest path problem on graphs with random arc costs which provides a so-called *policy* that leads from an origin to a destination node with minimal expected cost. Gao and Chabini (2006) further extend this concept into the *optimal routing policy* problem on dynamic stochastic networks, where link travel times are modelled as random variables with time-dependent distributions. Another valuable concept is the *hybrid route choice* model, first introduced by Mahmassani and Jayakrishnan (1991), where users can adapt their path while travelling. Waller and Ziliaskopoulos (2002) utilize this concept to introduce the *online shortest path* problem, where decision makers have the opportunity to reevaluate their remaining path en-trip when information becomes available.

Coupling en-trip path adaptation and stochastic arc costs leads to the concept of *adaptive routing policy*, defined as a decision rule that maps all possible revealed traffic conditions to the next link choices, subject to the available information (Gao et al., 2008). This allows to reproduce travellers' strategic behaviour when adapting to revealed traffic conditions en route. The random path utilities are sequentially updated at each node with the revealed traffic conditions and applied to revise the route choices. It is assumed that travellers know the probabilistic distributions of link travel times and the correlations among them. The computational burden of this assumption may somehow be reduced by: (1) an approximate heuristics (Gao and Chabini, 2006); (2) simplifying the unknown travel time distributions with incomplete statistics (Flajolet et al., 2017); or (3) sampling them from simulation results (Snowdon et al., 2012). Valuably, the adaptive routing policy can be further extended to the Prospect Theory (Gao et al., 2010), which allows to reproduce the twofold attitudes toward travel time variability: risk seeking under high probability and risk averse under low probability; that cannot be captured when the utility is assumed linear. Unnikrishnan and Waller (2009) proposed an equilibrium framework defined on routing policies instead of paths: the network is in equilibrium if no user can improve his/her expected cost by changing the routing policies (i.e. each user chooses a routing policy with minimum expected cost).

In contrast to the above, Hu and Mahmassani (1997) assume that users have access to full knowledge about the actual network costs. Perfectly informed users, however, behave with bounded rationality, so that rerouting takes place only if the difference between actual and typical travel times is greater than a given indifference threshold. In the model proposed by Mahmassani and Jayakrishnan (1991) drivers update their estimation of travel costs while travelling. The model was further extended through a rolling-horizon approach by Peeta and Mahmassani (1995), so that a given share of drivers periodically becomes aware about the actual travel costs and can update en-trip their path choice.

A similar hybrid model by Pel et al. (2009) requires path enumeration and is thus limited to a small networks, yet provides concepts which we adopt in this paper to model rerouting, namely: (1) the rerouting utility shall be computed subject to the destination; (2) drivers may perceive the received information in different ways and inaccurately. Corthout et al. (2011) complements this by stating that: (3) informed drivers can anticipate downstream congestion, and (4) the share of rerouting drivers is related to the travel cost gains from the current decision point to the destination. The other concepts founding our approach to rerouting are: (5) a specific version of multi-class assignment where informed and uninformed drivers are propagated jointly but utilizing different cost patterns (as proposed in Mahmassani and Jayakrishnan, 1991); (6) representing the awareness process as a transition of unaware drivers into aware drivers at intermediate nodes (Ma et al., 2016).

Another, lately popular approach (Dobler et al., 2012; Xiong et al., 2016; Ma et al., 2016; Cao et al., 2016) is to reproduce rerouting with agent-based microscopic simulators, which provide a flexible framework to model driver acquisition of information and path choice behaviour in detail, yet are by nature stochastic and complex.

The adaptive behaviour of drivers is broadly investigated in the day-to-day context. In particular, Cantarella and Cascetta (1995) define the probability of choosing a path in a given day as the product of: (a) the probability of reconsidering the choice done in the previous day and (b) the conditional probability of choosing that path given that the choice of the previous day is reconsidered. Watling and Hazelton (2003) simulate the learning process through an exponential smoothing: cyclically a share of users update the expected travel times with the actual experience. Han et al. (2011) analyse the impact of ATIS on day-to-day adaptation. This is further examined by Bifulco et al. (2016), showing that an inaccurate real-time information can destabilize the day-to-day dynamic system, and ATIS route indications are most efficient when the information is accurate. The day-to-day dynamic assignment is formulated by Bifulco et al. (2016) as a fixed point problem with predicted travel costs converging to the actual travel costs. In this paper we propose a similar fixed-point problem, but the DTA model is solved in the within-day context.

An excellent review of recent findings about the effects of information on travellers' behaviour can be found in Ben-Elia and Avineri (2015). The revealed behaviour is complex and includes several phenomena like reinforced learning, risk and loss aversion, probability weighting, anchoring and ambiguity aversion, regret aversion. These behavioural theories are supported by experimental findings. Selected results from Stated Preference experiments, where travellers choose a route in a specific information context, are: (1) prescriptive information, where drivers are explicitly informed with the optimal route, has the largest impact on behaviour (Ben-Elia et al., 2013); (2) real-time information availability at downstream nodes encourages participants to less risk-averse behaviour (Lu et al., 2010); (3) more detailed VMS information yield higher diversion rates (Peeta et al., 2000); (4) drivers' reaction is both heterogenic and stochastic (Dell'Orco and Marinelli, 2017); (5) providing information on the range of travel times, i.e. the difference between maximum and minimum, can trigger risky behaviour (Katsikopoulos et al., 2002). In the typical recent Revealed Preference experiments travellers are traced with GPS (Papinski et al., 2009; Schlaich, 2010; Vacca and Meloni, 2015). Ramos (2015) applied the sequential logit model of Fosgerau et al. (2013) on GPS traces and revealed that drivers: (1) are more sensitive to actual travel times when they are informed; (2) comply with pre-trip, rather than en-trip, information, but (3) consult travel information just before departing; (4) make distinctions among information sources which then impact differently on their behaviour.

The researches on Twitter data (e.g. Lerman and Ghosh, 2010; Sakaki et al., 2010; Bakshy et al., 2012) provide valuable insights on the information spread process. Its dynamics and range have been observed through the tweets posted after relevant emergencies: earthquake, hurricane, traffic events, rebellions, etc. Procter et al. (2013) show how information propagated through the Twitter network in time and obtained a Rayleigh-like shape of the spread curve, in-line with evacuation models (Lindell and Prater, 2007). Ghosh and Lerman (2014) show that information in the communication network is either completely negligible or significant and spreads like a virus, being reposted forward with exponential probability.

1.2. Objectives and contributions

Although the rerouting phenomenon was the object for many researches, it can be still considered an open issue, since the applicability of the proposed models is limited, due to excessive simplification or complexity.

Our contribution is a workable model that can be effectively applied on real-size networks to realistically represent the joint effects on supply (congestion) and demand (rerouting) of a single traffic event in presence of multiple information sources. To this end we extend the framework for macroscopic within-day DTA presented in Gentile (2016).

The methodology presented in this paper is built mainly on our previous research on DTA (Bellei et al., 2005; Bellei et al., 2006; Gentile et al., 2007; Gentile, 2015), which exploits implicit path enumeration through a sequential approach to route choice (Gentile and Papola, 2006; Long et al., 2013) and the Link Transmission Model (Yperman, 2007; Gentile, 2010; Nezamuddin and Boyles, 2015) as a simulation engine on the supply side.

After pointing out the limitations of the classic DTA to represent rerouting, we identify a suitable way to include the phenomenon in the model structure. Specifically, we based the proposed formulation on implicit path enumeration, as this allows to reproduce local en-trip decisions at nodes for users directed toward a given destination and to avoid the computation of shortest paths.

We adopted temporal layers to represent dynamics as this eases the flow handling when travel times change, which is instead challenging in trajectory based algorithms. Specifically, the path choices of the reference equilibrium can be used in the event scenario. Moreover, temporal layers allow for a less dense time discretization with long time intervals (e.g. 5–10 min), which is crucial for real-time applications and large networks.

Then, we present the ICM, which is suited for the specific DTA framework and reproduces rerouting as a two-stage process: first, drivers become aware about the event and its consequences on traffic; second, drivers may decide to change path.

We represent the probability of becoming aware of the event as the joint probability of being informed by various ATIS sources (radio, VMS, mobile apps), not only as a function of their penetration rate, but also of the space and time coordinates of the driver wrt those of the event, as in Kucharski and Gentile (2015).

We represent the probability of rerouting by aware drivers, who are reluctant to change, through a random utility model, which takes into account expected gains and avoided losses.

This way we are capable of representing the evolution of the rerouting phenomena in time and space when the information about the event and its consequences on traffic spread onto the network.

Since the outcomes of rerouting decisions are dependent on rerouting decisions made by other drivers, providing a sound equilibrium model is non-trivial. To this end we propose a fixed-point framework which converges to a solution where rerouting drivers will not only reduce their costs, but also their experienced travel costs will be in line with the expected ones.

Finally, ICM is the first (up to authors knowledge) macroscopic DTA model capable of simulating a scenario with one traffic event on large-size networks in a reasonable computing time (comparable to dynamic user equilibrium), while representing the effect of multiple information sources and stochastic rerouting decisions.

2. A macroscopic Dynamic Traffic Assignment model to road networks

2.1. Schematic formulation of the DTA model

The road network is here represented through a directed graph (N, A) , where N is the set of nodes and $A \subseteq N \times N$ is the set of arcs. Each arc $a \in A$ has a (possibly time varying) vector of supply characteristics $\mathbf{s}_a(\tau)$, e.g. length, free flow speed, capacity, jam density, signal settings, etc., that allow to represent its performances under different traffic conditions. The initial node of the generic arc $a \in A$, referred to as *tail*, is denoted $N_a^- \in N$, while the final node, referred to as *head*, is denoted $N_a^+ \in N$. The set of arcs exiting the generic node $i \in N$, referred to as *forward star*, is denoted $A_i^+ = \{a \in A: N_a^- = i\}$; the set of arcs entering node $i \in N$, referred to as *backward star*, is denoted $A_i^- = \{a \in A: N_a^+ = i\}$.

The subset of nodes where trips can start and/or end, referred to as *centroids*, is denoted $Z \subseteq N$. The travel demand of road users is given as a fixed (but time varying) flow $d_{od}(\tau)$ of (equivalent) vehicles departing at time τ from each origin $o \in Z$ and directed towards each destination $d \in Z$, which is also referred to as the *dynamic O-D matrix*.

Notation will follow some general rules. The subscripts of a variable are used to denote network elements, e.g. i for node and a for arc. Because the proposed assignment model is destination based, we make an exception and use superscript d to refer it. Since the model is dynamic, each variable is represented as a piecewise continuous function of the clock time τ , which is referred to as *temporal profile*; by omitting the dependence to τ we refer to the whole temporal profile. The variables of the proposed rerouting model often refer alternatively to the typical traffic conditions (denoted with a hat) of that day-type, or to the actual traffic conditions (denoted without a hat) which are a consequence of the simulated event.

Below we list the main variables and abbreviations used in the paper; all other variables are introduced as they appear.

$a \in A, i \in N$	generic arc and node of the graph
$o, d \in Z \subseteq N$	generic origin and destination centroids
$N_a^-, N_a^+ \in N$	tail and head of a generic arc $a \in A$
$A_i^-, A_i^+ \subseteq A$	backward and forward star of a generic node $i \in N$
$\mathbf{s}_a(\tau)$	supply characteristics of arc $a \in A$ at time τ
$q_i^d(\tau)$	flow exiting node $i \in N$ at time τ directed towards destination $d \in Z$
$q_a^d(\tau)$	flow entering arc $a \in A$ at time τ directed towards destination $d \in Z$
$d_{od}(\tau)$	demand flow from origin $o \in Z$ to destination $d \in Z$ at time τ .
$\hat{p}_a^d(\tau), \tilde{p}_a^d(\tau)$	arc conditional probability of entering arc $a \in A$ for drivers directed towards destination $d \in Z$ who exit from its tail N_a^- at time τ , actual and typical values
$t_a(\tau), \hat{t}_a(\tau)$	travel time of arc $a \in A$ for drivers entering it at time τ , actual and typical values
$\theta_a(\tau)$	exit time of arc $a \in A$ for drivers entering it at time τ
$\theta_a^{-1}(\tau)$	entry time of arc $a \in A$ for drivers exiting it at time τ
$c_a(\tau)$	travel cost of arc $a \in A$ for drivers entering it at time τ
$w_i^d(\tau)$	node potential, i.e. expected value of the minimum route cost perceived to reach destination $d \in Z$ by drivers exiting node $i \in N$ at time τ
$w_a^d(\tau)$	arc potential to reach destination $d \in Z$ by drivers entering arc $a \in A$ at time τ
Rerouting variables:	
$\iota_a(\tau)$	share of unaware drivers entering arc $a \in A$ at time τ who become aware (i.e. informed) about the event while traveling along the arc
$\kappa_i^d(\tau)$	share of aware drivers exiting node $i \in N$ at time τ directed towards destination $d \in Z$ who decide to reroute
$v_i^d(\tau)$	utility to reroute for drivers exiting node $i \in N$ at time τ and directed towards destination $d \in Z$
$\hat{q}_i^d(\tau), \tilde{q}_i^d(\tau), \tilde{\tilde{q}}_i^d(\tau)$	unaware, aware and rerouted flow of drivers exiting node $i \in N$ at time τ directed towards destination $d \in Z$
Abbreviations:	
DTA	Dynamic Traffic Assignment
ICM	Information Comply Model
RCM	Route Choice Model
FPM	Flow Propagation Model
MSA	Method of Successive Averages
ACM	Arc Cost Model
APM	Arc Potential Model
LRM	Local Routing Model
EAM	Event Awareness Model
ERM	Event Routing Model

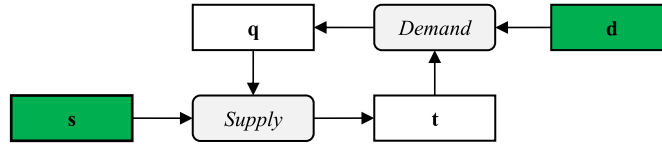


Fig. 2. Simplified schema of the Dynamic Traffic Assignment model.

Below, we describe in a compact form the DTA model to which the proposed ICM was applied; for more details we refer to Gentile (2016).

Dynamic Traffic Assignment can be regarded as the system of two sub-models: *supply*, being a function of the demand, and *demand*, being a function of the supply.

The supply model yields the network performance pattern in terms of travel times $\mathbf{t}$ corresponding to the network usage pattern in terms of destination flows $\mathbf{q}$ for given supply characteristics $\mathbf{s}$:

$$\mathbf{t} = \text{Supply}(\mathbf{q}; \mathbf{s}). \quad (1)$$

The proposed DTA considers a macroscopic traffic flow model based on the kinematic wave theory, capable to reproduce the main traffic phenomena (hypocritical congestion, queues and spillback). The supply model is crucial for DTA, but is not the focus of this paper; for details on how network performances are simulated in the proposed model refer to Gentile (2015). However, in the ICM it can be easily substituted with any alternative simulation model of traffic flows.

The demand model yields the network usage pattern in terms of destination flows $\mathbf{q}$ corresponding to the network performance pattern in terms of travel times $\mathbf{t}$ for a given demand o-d matrix $\mathbf{d}$:

$$\mathbf{q} = \text{Demand}(\mathbf{t}; \mathbf{d}). \quad (2)$$

Travel costs $\mathbf{c}$ of users are first obtained by combining the results of the traffic flow model (i.e. travel time, queue length, etc.) with other relevant (flow-independent) attributes considered by drivers when making their route-choice (i.e. safety, toll, etc.). Then, a sequential (Gentile and Papola, 2006) Route Choice Model (RCM), deterministic or stochastic, allowing for implicit path enumeration (Gentile and Noekel, 2016) is applied to obtain the arc potentials $\mathbf{w}$ and the arc conditional probabilities $\mathbf{p}$, for each destination, like in Dial (1971). Finally, a Flow Propagation Model (FPM) is applied to load travel demand on the network, based on the arc conditional probabilities, thus obtaining the destination flows $\mathbf{q}$. Both the RCM and the Flow FPM require the travel times $\mathbf{t}$, being this the main difference from static and dynamic assignment models.

The combination of the two sub-models yields a fixed point problem (Bellei et al., 2005; Gentile et al., 2007) which formalizes the user equilibrium, i.e. a traffic condition where no driver can improve his (perceived, in the case of stochastic route choice model) path cost.

The heuristic method to approximate the fixed-point is iterative applying following Method of Successive Averages (MSA), where the step size α_n at iteration n is usually set equal to $1/n$:

$$\mathbf{q}^n = (1 - \alpha_n) \cdot \mathbf{q}^{n-1} + \alpha_n \cdot \text{Demand}(\text{Supply}(\mathbf{q}^{n-1}; \mathbf{s}); \mathbf{d}). \quad (3)$$

Faster and more exact methods than MSA are today available (e.g. Gentile, 2016), although the former is still widely used heuristic thanks to its robustness and generality, which is paid with slow convergence.

In general, the DTA model provides the equilibrium temporal profiles of the main traffic variables (travel times $\mathbf{t}$ and destination flows $\mathbf{q}$, from which it is immediate to obtain arc costs, volumes and probabilities, if needed) for a given supply $\mathbf{s}$ and demand $\mathbf{d}$ (schematically depicted at Fig. 2):

$$(\mathbf{t}, \mathbf{q}) = \text{DTA}(\mathbf{s}, \mathbf{d}). \quad (4)$$

2.2. Focus on route choice and flow propagation

Since the rerouting phenomenon takes place at the demand side of DTA, in the following we present this model in some more detail.

Arc travel times play here three roles:

1. they are a major component of the arc costs perceived by drivers;
2. they are used to evaluate the arc costs at the time when the driver enters them;
3. they are used to propagate the flows on the network.

Concerning point 1, the arc costs are assumed to depend linearly on the travel times through the value of time γ and a constant (monetary) term $\chi_a(\tau)$, which summarizes all flow-independent components of disutility:

$$c_a(\tau) = \gamma \cdot t_a(\tau) + \chi_a(\tau). \quad (5)$$

Eq. (5) can be expressed in compact form as the Arc Cost Model:

$$\mathbf{c} = \text{ACM}(\mathbf{t}). \quad (6)$$

Concerning points 2 and 3, it is useful to introduce the exit time and its inverse, which expresses the entry time, as a function of the travel time:

$$\theta_a(\tau) = \tau + t_a(\tau), \quad (7)$$

$$\theta_a^{-1}(\tau + t_a(\tau)) = \tau. \quad (8)$$

Obviously, the above formulation is generic, and state-of-the-art traffic flow models are more complete. Specifically, to ease the presentation of the model, in the following we assume that the temporal discretization used for the numerical solution is sufficiently dense, so that no vehicle can travel any arc in less than one time interval. In practical implementations this assumption is relaxed, because longer time intervals lead to less variables and then shorter computation runtime (Gentile et al., 2007; Himpe et al., 2016); but this implies a slightly more complex model where the acyclicity of the corresponding space-time network does not hold anymore. Likewise, in practical implementations handle empty arcs at the beginning and end of the simulation period, flat cumulative flow profiles for periods with no inflow, etc.

The proposed demand model with implicit path enumeration requires to process the nodes twice. First, node potentials $w_i^d(\tau)$ are determined recursively in reverse chronological order (but in no particular topological order) as a function of later node potentials of the forward star:

$$w_i^d(\tau) = w(w_a^d(\tau) : a \in A_i^+), \quad (9)$$

$$w_a^d(\tau) = c_a(\tau) + w_{N_a^+}^d(\theta_a(\tau)). \quad (10)$$

In Eq. (10) the arc potential $w_a^d(\tau)$ is introduced as the sum of the arc cost at time τ and of its head node potential evaluated at the (later) exit time $\theta_a^d(\tau) > \tau$. The specific function w to apply in Eq. (9) depends on the behavioural assumption made for route choice: e.g. deterministic models would apply a Bellman relation (like in Gentile, 2016):

$$w_i^d(\tau) = \text{Min}(w_a^d(\tau) : a \in A_i^+) \quad (11)$$

while Logit models would apply a logsum formula (like in Bellei et al., 2005), where the parameter ϑ is proportional to the standard deviation of the random perception error:

$$w_i^d(\tau) = -\vartheta \cdot \text{Log} \left(\sum_{a \in A_i^+} \text{Exp} \left(-\frac{w_a^d(\tau)}{\vartheta} \right) \right). \quad (12)$$

The arc potentials can then be expressed in compact form through the Arc Potential Model as a function of arc performance:

$$\mathbf{w} = \text{APM}(\mathbf{c}, \mathbf{t}). \quad (13)$$

Then, the conditional probabilities $p_a^d(\tau)$ of each arc $a \in A_i^+$ exiting node $i \in N$ are obtained, in any order, as a vector function of the arc potentials of the forward star.

$$\mathbf{p}_{A_i^+}^d(\tau) = p(w_b^d(\tau) : b \in A_i^+). \quad (14)$$

Again, the specific function p to apply in Eq. (14) depends on the behavioural assumption made for route choice: e.g. deterministic models would apply Wardrop's principle, so that the arc conditional probabilities must jointly satisfy, besides consistency constraints (i.e. non negative values summing up to one), the following complementarity conditions (like in Gentile, 2016):

$$p_a^d(\tau) \cdot (w_a^d(\tau) - w_i^d(\tau)) = 0 \quad \forall a \in A_i^+ \quad (15)$$

while Logit models would apply a negative exponential proportion (like in Bellei et al., 2005):

$$p_a^d(\tau) = \text{Exp} \left(-\frac{w_a^d(\tau) - w_i^d(\tau)}{\vartheta} \right). \quad (16)$$

The arc conditional probabilities can then be expressed in compact form through the Local Routing Model as a function of arc potentials:

$$\mathbf{p} = \text{LRM}(\mathbf{w}). \quad (17)$$

The Route Choice Model expresses in compact form the Eqs. (9), (10) and (14):

$$\mathbf{p} = \text{LRM}(\text{APM}(\mathbf{c}, \mathbf{t})) = \text{RCM}(\mathbf{c}, \mathbf{t}). \quad (18)$$

Although in principle route choice can be modelled equivalently through a deterministic or a stochastic model, the presence in the proposed framework of other probabilistic events (being informed and deciding to reroute) would induce to opt for the second approach.

Subsequently, the node flows $q_i^d(\tau)$ by destination, together with the flows $q_a^d(\tau)$ of each arc $a \in A_i^+$ exiting node $i \in N$, are obtained recursively in direct chronological order (but in no particular topological order) based on:

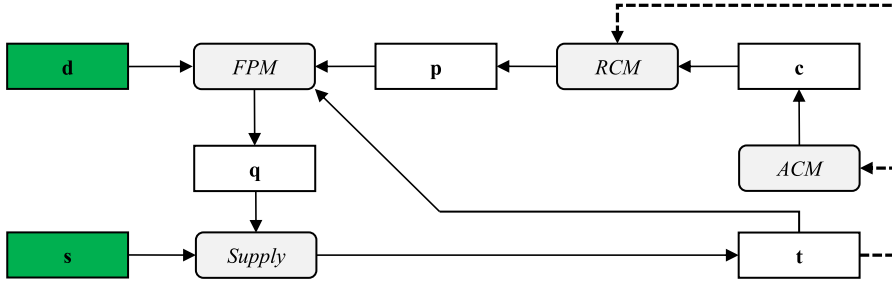


Fig. 3. Schema of the Dynamic Traffic Assignment model with detailed demand model.

1. flow conservation, which involves the o-d demand matrix and the earlier flows entering the arcs of the backward star;
2. temporal consistency, which involves the entry time derivative, because speed variations imply vehicle dispersion or concentration:

$$q_i^d(\tau) = d_{id}(\tau) + \sum_{a \in A_i^-} q_a^d(\theta_a^{-1}(\tau)) \cdot \frac{\partial \theta_a^{-1}(\tau)}{\partial \tau}, \quad (19)$$

$$q_a^d(\tau) = q_{N_a}^d(\tau) \cdot p_a^d(\tau). \quad (20)$$

In Eq. (20) the arc flow $q_a^d(\tau)$ is simply obtained by multiplying the node flow and the arc conditional probability.

Note that in practice the arc conditional probabilities can be computed together with the arc potentials or immediately before the destination flows; therefore demand model requires to process nodes only twice.

The FPM expresses in compact form the Eqs. (19) and (20):

$$\mathbf{q} = \text{FPM}(\mathbf{p}, \mathbf{t}; \mathbf{d}). \quad (21)$$

Finally the demand model is the composition of ACM, RCM, and FPM:

$$\text{Demand}(\mathbf{t}; \mathbf{d}) = \text{FPM}(\text{RCM}(\text{ACM}(\mathbf{t}), \mathbf{t}), \mathbf{t}; \mathbf{d}). \quad (22)$$

To obtain the whole flow pattern, each destination $d \in Z$ is processed in any order. Then, the destination flows of each arc $a \in A$ are usually aggregated into the link volume $q_a(\tau)$ of vehicle equivalents. Indeed, besides this being a fundamental output of DTA together with travel times, volumes suffice to source simple point queue models, while destination flows are required only by more detailed supply models which are though capable of representing spillback phenomena (Gentile, 2015; Himpe et al., 2016).

If in the above schema (Fig. 3) we exclude the dashed arrows (i.e. we assume given probabilities) the resulting model is a Dynamic Network Loading. At this point it is worth noting how the demand – supply system is conceptually different from the classical decomposition of DTA into a Dynamic Network Loading model and a RCM. The latter resembles more how a user equilibrium may be reached in reality as a day-to-day learning and updating process: drivers make their route decision, possibly updating the choice of the previous day, based on an estimation of performance pattern in which the travel times of the previous day are part of a learning mechanism that takes into account also all previous experiences (and, possibly, pre-trip information). Instead, the demand model does not correspond to a process that takes place in reality, and makes sense only as a component of the equilibrium together with the supply model.

2.3. Numerical solution of the demand model

In the numerical implementation of the method (Algorithm 1) we assume a piecewise approximation where each variable is computed only at predefined instants τ_t , with $t \in T = [0, \mu]$. In particular, performance variables (like travel times and arc potentials) are assumed piecewise linear while usage variables (like destination flows and arc probabilities) are assumed piecewise constant in the interval $\tau_t \leq \tau < \tau_{t+1}$, with $0 \leq t < \mu$; the last μ th interval represents all instants $t \geq \tau_\mu$.

Temporal consistency is taken into account, not through the exit and entry time, but by:

- the *entry-exit map*, with $e_{at} \in T$ being the exit interval of a vehicle entered at time τ_t with $t \in T$, i.e. $\tau_e \leq \tau_t + t_{at} < \tau_{e+1}$, with $e = e_{at} > t$ (by assumption);
- the *propagation map*, with m_{ate} being the share of flow entering arc a during interval $t \in T$ that exit it during interval $e \in T > t$ – to reduce memory consumption, in the implementation only the positive elements of the map are instanced.

Both maps are a result of the supply model, together with the travel times t_{at} as all such variables can be obtained from the temporal profiles of the exit times:

$$m_{ate} = \frac{\text{Min}(\text{Min}(\theta_a^{-1}(\tau_{e+1}), \tau_{t+1}) - \text{Max}(\theta_a^{-1}(\tau_e), \tau_t), 0)}{\tau_{t+1} - \tau_t}. \quad (23)$$

Algorithm 1 Demand model.

```

function Demand(t, e, m; d)                                * for given travel times and od matrix
   $c_{at} \leftarrow ACM(t_{at}) \quad \forall a \in A, \forall t \in T$                 * compute travel costs
  for each  $d \in Z$                                                 * for each destination
    

---


    first node visit in reverse chronological order
     $w_{N_{\mu}}^d \leftarrow ShortestTree(c_{A_{\mu}}, d)$                 * compute minimum node costs at the last instant
    for  $t = \mu-1$  to 0                                          * for each instant in reverse chronology
      for each  $i \in N$                                             * process each node in any order
        for each  $a \in A_i^+$                                        * for each arc of the forward star
           $j \leftarrow N_a^+$                                        * consider the head node
           $e \leftarrow e_{at}$                                        * consider the exit interval

           $w_{at}^d \leftarrow c_{at} + w_{je}^d + (\tau_i + t_{at} - \tau_e) \cdot \frac{w_{je+1}^d - w_{je}^d}{\tau_{e+1} - \tau_e}$  * arc potential at the exit time

        next  $a$ 
         $w_{it}^d \leftarrow w_{at}^d : a \in A_i^+$                         * node potential for the given choice model
      next  $i$ 
    next  $t$ 
    

---


    second node visit in chronological order
     $q_{NT}^d \leftarrow 0_{NT}$                                        * reset all node flows
    for  $t = 0$  to  $\mu-1$                                           * for each interval in chronology
      for each  $i \in N$                                             * process each node in any order
         $p_{A_i^+}^d \leftarrow p(w_{bt}^d : b \in A_i^+)$                 * arc probabilities for the given choice model
        if  $i \in Z$  then  $q_{it}^d \leftarrow q_{it}^d + d_{idt}$           * load the od demand on the node if an origin
        for each  $a \in A_i^+$                                        * for each arc of the forward star
           $q_{at}^d \leftarrow q_{it}^d \cdot p_{at}^d$                        * compute the flow
           $j \leftarrow N_a^+$                                        * consider the head node
          for  $e = e_{at}$  to  $e_{at+1}$                                 * for each exiting interval
             $q_{je}^d \leftarrow q_{je}^d + q_{at}^d \cdot m_{ate}$           * propagate the arc flow at the head node
          next  $e$ 
        next  $a$ 
      next  $i$ 
    next  $t$ 
    

---


    next  $d$ 
    return  $q$                                                     * return destination flows
  end function

```

We assume that all users complete their trip before the end of the simulation, so that the travel times after the end of the simulation are uncongested.

2.4. DTA model limitations and potential to represent rerouting phenomena

Neither DTA, nor DNL as formulated above are suitable to simulate unexpected events and in particular to model rerouting phenomena. Indeed, in DTA the RCM assumes predictive information, and in DNL fixed demand pattern used in the propagation model.

These are two extremal scenarios; typically part of the drivers behave accordingly with the DTA principle, and part with the DNL. In the present paper we aim at dealing with such mixture of conditions.

Let us first consider the limit of DTA. In a dynamic RCM decisions are made based on future travel times taken from previous experience as well as from traffic information and forecast. Implicitly, such predictive route choice paradigm implies that a driver departing from his origin knows in advance the travel times that he will encounter until he arrives at his destination: in the proposed model this is attained with Eq. (10). Using the actual network characteristics including the event to solve DTA results then in full knowledge about the future traffic states. Consequently, routing decisions would be made to avoid the event before it occurs or becomes known, leading to unrealistic patterns.

Now, let us consider the limit of DNL. In the FPM the demand flows are loaded using routing decisions made pre-trip, based on expected network performances not including the event. Thus, if the actual travel times happen to be different from what was expected, only the propagation will be affected but no rerouting can be simulated.

The proposed ICM model was conceived to extend the above model structure of DTA to simulate rerouting phenomena. Namely it exploits:

1. implicit path enumeration, where route choice probabilities are reproduced as splitting rates from any node to each destination;
2. supply-demand separation, where actual travel times can be obtained for flows propagated with several independent route choice models;
3. strong distinction between travel times and perceived travel costs, which allows to propagate flows along the path of minimal perceived costs (possibly outdated) consistently with actual travel times;
4. flow conservation thanks to temporal layers formulation with propagation maps.

Thanks to these features, the ICM overcomes the above limits by ensuring that actual travel times cannot be perceived by unaware drivers in their route choices, while all drivers, not just those rerouting, are propagated on the basis of the actual travel times.

3. The Information Comply Model

Rerouting takes place when a driver tries to escape from some traffic event and en-trip decides to change path. We assume the following rerouting behaviour:

1. a driver reroutes only once he becomes aware of the event, which may occur thanks to several information sources, including direct observation;
2. the reason for rerouting is to avoid the negative consequences of the event and benefit from expected lower costs, which shall then more than compensate his reluctance to change;
3. the driver reroutes to an optimal path considering the correct estimation of actual and future travel costs produced by the event and by the rerouting of other drivers, based on experiences of similar situations and/or on predictive traffic information.

This behaviour is reproduced in the ICM as a sequence of states and changes in the mental condition of a driver who:

- 1. is initially unaware about the event and follows his habitual path from his origin toward his destination, which has been chosen pre-trip on the basis of the typical within-day evolution of travel times on the network in that day-type;
- 1–2. becomes informed about the event, while travelling on a certain arc of his habitual path, by some source of traffic broadcast and/or by observing the current situation: he can use his own eyes to spot queues, or he can experience anomalous travel times along his route;
- 2. is now aware of the event (and, by hypothesis, of all its consequences, including other drivers rerouting) but does not see yet the necessity to reroute, although at every node he evaluates this alternative by properly considering the relative advantage of changing path, by comparing the actual and typical travel times, as well as the stress of doing it;
- 2–3. decides to reroute at a certain node of his habitual path, when he reaches the conclusion that estimated delays are so high that the situation deserves action to save a consistent amount of time;
- 3. follows the new optimal path until the destination, because his experience and/or available predictive information allow him to make a correct forecast of the traffic that he will encounter from now on along the trip without major surprises.

We have then assumed (and broadly discuss in [Section 3.3](#)) that:

- when the driver is informed, he perfectly becomes aware of the actual travel times, i.e. he is able to figure out the exact nature of the event;
- experienced and/or informed drivers can predict, not only the direct consequences of the event (e.g. queues upstream of the event and their spillback), but also that other drivers will reroute and generate congestion on alternative paths.

Clearly, the above strong assumptions represent a schematic simplification which is useful to model a complex phenomenon like rerouting, whereas in reality knowledge of drivers is never perfect and information systems are never fully accurate and predictive. However, these factors can be partially included in the proposed framework by considering a stochastic route choice model (like the multinomial logit) rather than a deterministic one.

The demand flow in ICM is propagated in the three subsequent *conditions*: *unaware*, *aware* and *rerouted*; with two transitions: *information*, i.e. becoming aware (ICM *awareness* model), and *comply*, i.e. deciding to reroute (ICM *rerouting* model).

We consider the user equilibrium results of DTA computed on the network without event as the *typical* traffic conditions, denoted with a hat.

The immediate effects of the event can regard both the supply characteristics $\mathbf{s}$ (usually a temporary modification of some arc capacity or free flow speed) and the dynamic o-d matrix $\mathbf{d}$ (usually an impulse in the generation or attraction of some centroids). ICM allows to simulate one single event, which however can consist of a connected set of supply and/or demand changes (e.g. some roads are closed for the Christmas market and the evening demand of the area is increased; links are closed dynamically as a marathon moves through the city). As long as the information of all these changes spreads jointly among drivers, the assumptions of the ICM hold fine. On the contrary, simulating a sequence of mostly independent

events (e.g. one accident on the motorway and another one inside the city an hour later) is impossible with the ICM and requires, for example, a rolling horizon approach, like in Gentile et al. (2013), who proposed a much simpler but more flexible rerouting model.

By including these events in the DTA model and by replacing the classical demand sub-model with the above described ICM (only in a qualitative way, so far), we obtain the *actual* traffic conditions, denoted without a hat. Therefore, typical conditions are fixed and precomputed, while actual conditions are the result of a new user equilibrium with rerouting, which will be found only when convergence of the proposed DTA method is attained.

3.1. The awareness model

The awareness model aims at representing how drivers become aware of the unexpected event. We use the term *aware* to denote both being informed about the event and becoming conscious of experiencing atypical delays. The proposed model shall include not only all the major sources of traffic information but also the educated guessing of experienced drivers.

As already signalized in the introduction, there are numerous ways the traffic information can be provided to the drivers. For a review of fundamental ATIS strategies and information types see Khattak et al. (1993). Here we consider and unify three main information types: (a) generic (e.g. “tunnel blocked”), (b) prescriptive (e.g. “tunnel blocked, to reach City use A4”), (c) personalized (e.g. “tunnel blocked, to reach destination follow this path”). We argue that they can be unified in one cognitive paradigm for the rerouting driver. Namely, we assume that the aware driver (regardless what type of information he received) will act as if he is aware of the actual network travel times and costs and their future evolution resulting from the event and the drivers reaction to this event (rerouting), as further discussed in Section 3.3.

Specifically, we assume that the most accurate forecast of travel costs is available at the level of Traffic Management Centre (TMC) and broadcasted as a real-time route guidance, optimized VMS message, or online query made by drivers curious of the reason for the experienced delay. If, however, the driver autonomously updates his expectations on travel costs, it must be assumed that the driver is able to correctly predict the actual network costs. Under the above assumptions, we unify the information provision types and assume that drivers become aware of the current and future network costs. This allows to handle multiple information sources jointly and to propose a single behavioural model for rerouting (introduced in Section 3.2). Though, including the consequences of other drivers rerouting in the prediction of the traffic pattern requires to solve a circularity introduced in Section 3.3.

Awareness in ICM is treated as a binary variable: the driver is not aware as long as he did not receive information from any of the broadcasting sources or he did not get conscious of the event by its own means. The probability $\iota_a(\tau)$ of becoming aware while traversing arc $a \in A$ and exiting it at time τ depends on the probability $\iota_a^s(\tau)$ of becoming aware by means of each way $s \in S$ among a set of available information sources. The probability of becoming aware from at least one source can be expressed as the complementary event that the driver does not become aware by any source. To make the model practical, we assume that all sources are mutually independent, so that the joint probability $\iota_a(\tau)$ can be expressed as the product of the complementary probabilities for all available sources $s \in S$:

$$\iota_a(\tau) = 1 - \prod_{s \in S} (1 - \iota_a^s(\tau)). \quad (24)$$

The above general concept is designed to work with the most popular sources of traffic awareness that are present in contemporary road networks (Kucharski and Gentile, 2016). In the following, we propose models for the main paradigms of information spreading:

1. fixed time information, e.g. news broadcasted through the radio or cyclical TMC updates;
2. fixed place information, e.g. messages shown at a road-side VMS;
3. on-line information, e.g. published on a webpage, or via mobile applications;
4. experiencing atypical traffic conditions, e.g. spotting unexpected queueing.

3.1.1. Fixed time information

Information like radio news and TMC traffic data are cyclically broadcasted and can reach a share of recipient drivers π_{NEWS} (market penetration) wherever they are. At each subsequent broadcast time τ_n^{NEWS} , with $n = 1, 2, \dots$, only a portion π_{ON} of unaware drivers who are actually listening to the news become aware, regardless their location.

The total share $I^{NEWS}(\tau)$ of informed drivers increases with every broadcast i , adding the probability that the driver was not informed by previous broadcasts (equal to the power $i-1$ of $1-\pi_{ON}$) and is informed by that broadcast (equal to π_{ON}); it asymptotically reaches the market penetration.

$$I^{NEWS}(\tau) = \pi_{NEWS} \cdot \pi_{ON} \cdot \sum_{i=1}^{\text{Max}(n: \tau \leq \tau_n^{NEWS})} (1 - \pi_{ON})^{i-1}. \quad (25)$$

The probability of becoming aware through the radio news $\iota_a^{NEWS}(\tau)$ is a function of how many times the news are broadcasted while the driver is travelling along arc $a \in A$; it can be obtained as the difference between the above cumulative probability at the exit and entry times:

$$\iota_a^{NEWS}(\tau) = I^{NEWS}(\tau + t_a(\tau)) - I^{NEWS}(\tau). \quad (26)$$

Mind that drivers can become aware via radio and other on-line sources also pre-trip, i.e. at their origin, before departure. To represent this circumstance a dummy arc is introduced at each origin from which users exit at the time of departure and enter at time $\tau = 0$, so that their aware share is correctly initialized. Moreover, part of the broadcast times can be earlier than the event time. With these expedients we can also represent the effect of predictable or known events of which drivers can be informed in advance.

3.1.2. Fixed place information

The information broadcasted at fixed locations, e.g. on Variable Message Signs (VMS), reaches only those drivers who traverse an arc equipped with a VMS while the latter is displaying the information about the event. The probability $\iota_a^{VMS}(\tau)$ that the event is considered by the monitoring system to be the most relevant for users passing arc a at time τ and is then displayed on its VMS (if any), hence making drivers aware, is here assumed to be a decreasing function of its total distance $\ell_a^e(\tau)$ from the event.

$$\iota_a^{VMS}(\tau) = \delta_a^{VMS} \cdot B(\tau \geq \tau^{VMS}) \cdot \frac{1}{1 + \left(\frac{\ell_{ae}(\tau)}{\ell_2}\right)^{\beta^{VMS}}}, \quad (27)$$

$$\ell_a^e(\tau) = \|a, e\| + \bar{v} \cdot \|Mid(0, \tau_0^e - \tau, \tau_1^e - \tau)\|, \quad (28)$$

where:

- δ_a^{VMS} is 1 if arc a is equipped with a VMS and is 0 otherwise;
- τ^{VMS} is the time when the information about the event enters the monitoring system;
- $B(x)$ is 1 if x is true and is 0 if x is false (Boolean function);
- $\|a, e\|$ denotes a Cartesian distance from the arc tail to the event;
- $Mid(a, b, c)$ is the middle value of a, b, c (used to evaluate the temporal distance);
- ℓ_2 is the total distance where the probability is halved (depends on the event relevance);
- β^{VMS} is a shape parameter to be calibrated;
- (τ_0^e, τ_1^e) are the temporal coordinates of the event (start time and end time);
- $\bar{v}$ is the average direct speed on the network to transform temporal into spatial distances.

In particular, the probability that the event is displayed on the VMS is assumed to have a hyperbolic decay wrt the power β^{VMS} of the total (spatial and temporal) distance: if the total distance of the event from the arc tail (where we suppose the VMS is placed) is null and the event is occurring now, then the probability is one; if the total distance from the event is ℓ_2 , then the probability is halved. Note that, after the end of the event it is still relevant to inform drivers of its consequences (i.e. queues may still be ongoing).

This isotropic hypothesis is valid only as a first approximation that may be accepted for off-line simulations. It does not take into account the actual topology of the network and the direction of demand flows. The proposed metric can be adjusted to reflect these aspects, like in the model used for the Virtual Message Sign inside OPTIMA (PTV SISTeMA, 2016), where a priority to appear on the screen is assigned for several concurrent events. Moreover, in the real time environment, instead of the above formula, a simple Boolean signal can be obtained directly from the traffic management system.

3.1.3. On-line sources

Equally interesting and challenging is to represent the traffic information available on-line, which can be accessed by drivers at any time. Such information sources are widely used nowadays, especially by habitual drivers, and include: mobile applications, social networks, online services, etc. Obviously browsing for information online while driving is hazardous as it distracts the attention off the road. Anyhow we presume that many people would still do it (e.g. while stuck in traffic). Moreover, considering that drivers interactions with apps is constantly improving drivers may receive information without the need to browse (e.g. voice control, or pop-up notifications), which also fits the below model.

Let's assume that the information about the event is first published on-line at time τ_{ONLINE} after which drivers utilizing the above type of sources with market penetration π_{ONLINE} may receive it. Drivers may notice the information while browsing, or because some of their contacts in a virtual community posted it. The cumulated probability of being informed gradually increases from zero at the publishing time to the total market penetration while the post spreads across the relevant social or media network.

We derive the on-line spreading profile from empirical findings relative to Twitter data. Procter et al. (2013) studied how tweets are re-tweeted and reach the new recipients. The revealed spreading process supported the expectations of early researchers that the Rayleigh distribution is suitable to estimate the probability of receiving the information at a given time τ after its first publication τ_{ONLINE} . Fig. 6 shows when the twitter information reached the recipients compared with the theoretical Rayleigh distribution.

The cumulative distribution of Rayleigh, whose only parameter is the spreading pace σ , multiplied by the market penetration yields then the probability $I^{ONLINE}(\tau)$ that a recipient driver is informed before time $\tau \geq \tau_{ONLINE}$:

$$I^{ONLINE}(\tau) = \pi_{ONLINE} \cdot \left(1 - \exp\left(-\frac{(\tau - \tau_{ONLINE})^2}{2 \cdot \sigma^2}\right)\right). \quad (29)$$

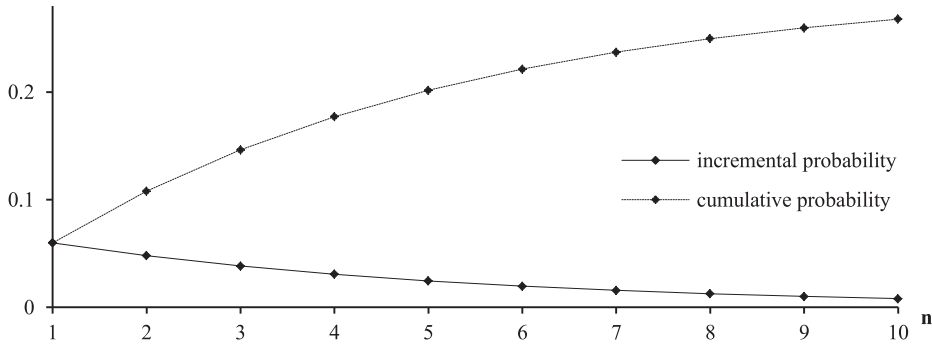


Fig. 4. Incremental and cumulative probability that drivers become aware at the n th news broadcast, for $\pi_{NEWS} = 30\%$ and $\pi_{ON} = 20\%$.

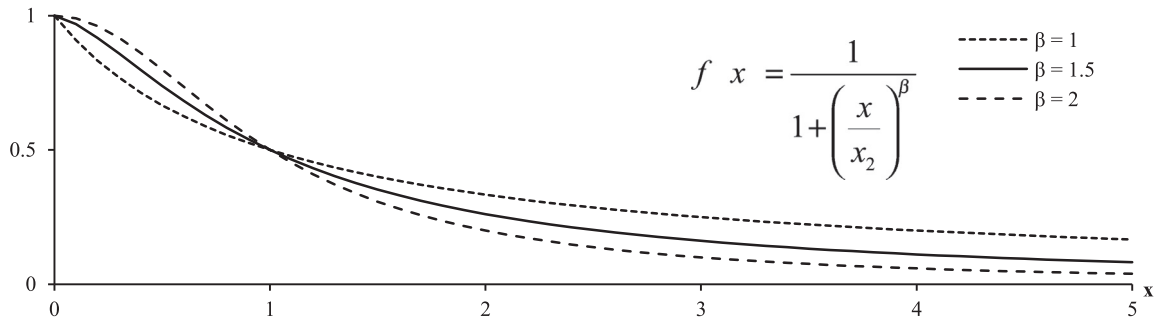


Fig. 5. Hyperbolic expression of the decreasing probability that a VMS displays the event as a function of its distance x , for a halving factor $x_2 = 1$ and different values of the power β .

The probability of becoming aware via an on-line source while traversing arc a is given by the difference between the above cumulative distribution at the exit and entry times:

$$I_a^{ONLINE}(\tau) = I^{ONLINE}(\tau + t_a(\tau)) - I^{ONLINE}(\tau). \quad (30)$$

On top of that, viral information reaches more recipients and reaches them faster (Ghosh and Lerman, 2014). In the following, we extend the above model with information *virality*. Although the process of becoming viral is highly random and there are no efficient methods to predict it (Hoang et al., 2011), some monitoring techniques are capable to identify if the information is spreading virally (Sakaki et al., 2010).

We assume that virality corresponds to the event significance, which can be measured with its impact on the total travel time. This is expressed in the following as the relative increase of the travel times spent on the network by all users (entered on arcs) up to the current time τ if the arc flows remained constant (we assume that arc closures can only be temporary so that the arc travel times are not infinite):

$$\Delta t(\tau) = \frac{\sum_{a \in A} \int_0^\tau t_a(\theta) \cdot \hat{q}_a(\theta) \cdot d\theta}{\sum_{a \in A} \int_0^\tau \hat{t}_a(\theta) \cdot \hat{q}_a(\theta) \cdot d\theta} - 1. \quad (31)$$

The complement $1 - \pi_{ONLINE}$ of the market penetration and the spreading pace σ of the Rayleigh distribution in Eq. (29) are then assumed to present a hyperbolic decay (see Fig. 4) wrt the relative increment $\Delta t(\tau)$:

$$\pi_{ONLINE}(\tau) = 1 - \frac{1 - \pi_0^{ONLINE}}{1 + \left(\frac{\Delta t(\tau)}{\Delta t_2}\right)^{\beta_{ONLINE}}}, \quad (32)$$

$$\sigma(\tau) = \frac{\sigma_0}{1 + \left(\frac{\Delta t(\tau)}{\Delta t_2}\right)^{\beta_{ONLINE}}}. \quad (33)$$

The initial values of the market penetration π_0^{ONLINE} and of the spreading pace σ_0 as well as the halving increment Δt_2 and the power β_{ONLINE} shall be calibrated to match field data (Mathioudakis and Koudas, 2010). Fig. 7 shows how the cumulative probability $I^{ONLINE}(\tau)$ changes for the viral events: the curve goes up and its banding is anticipated.

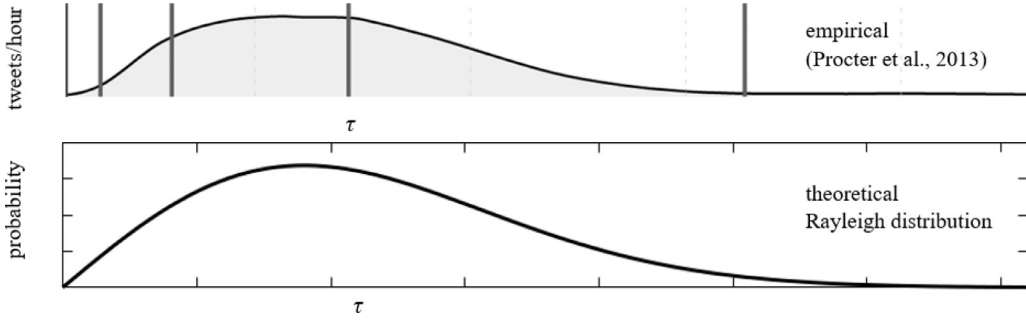


Fig. 6. Empirically observed frequency of tweets after an event and Rayleigh distribution.

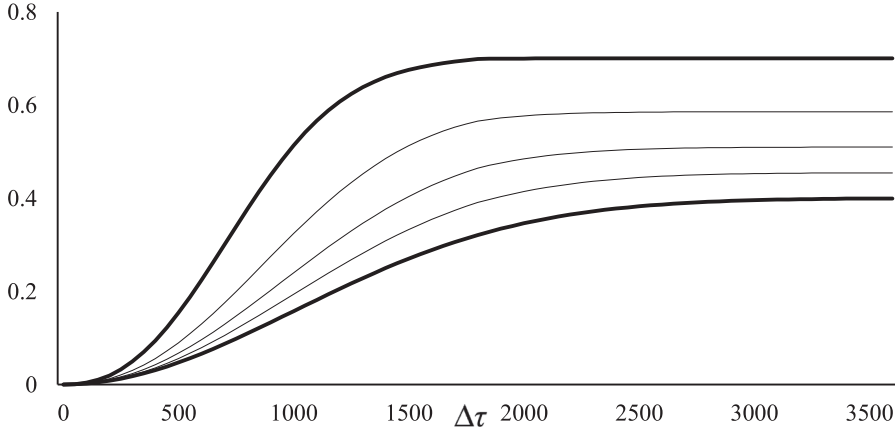


Fig. 7. Probability that a driver is informed within $\Delta\tau$ seconds after the event for a base penetration rate $\pi_0^{ONLINE} = 0.4$, assuming that $\Delta t / \Delta t_2$ is zero (bottom line), one (top line) and in between (thin lines); $\beta^{ONLINE} = 0.5$, $\Delta t_2 = 1800$ s.

3.1.4. Awareness through observation

Besides receiving information, drivers can become aware through observation. If an expert user of the road network experiences a relevant delay wrt typical travel times while travelling along an arc of his habitual path, he may correctly guess the reason and become aware about the event.

The motivation to assume this skilled behaviour is twofold. First of all, a driver is curious of the delay and may use the available information sources to understand its cause, e.g. he can search on-line. Secondly, experienced drivers can correctly estimate the network travel times by observing atypical travel times along their route (Gao et al., 2008), especially for recurrent but unexpected events (e.g. gridlocks in critical points of the network, or black-spots where accidents occur frequently).

The awareness probability $\iota_a^{OBS}(\tau)$ derived from observation is here assumed proportional to the delay experienced by the driver entering on arc a at time τ , i.e. the difference between actual and typical travel times, with a given threshold t_0^{OBS} and sensitivity ξ^{OBS} . A complete empirical evidence of this phenomenon is yet missing; more realistic formulations could be devised as revealed field data will be available (e.g. trajectories of floating car data).

$$\iota_a^{OBS}(\tau) = \text{Mid}(0, 1, \xi^{OBS} \cdot (t_a(\tau) - \hat{t}_a(\tau)) - t_0^{OBS}). \quad (34)$$

The proposed model assumes each arc as an independent source of awareness through observation, which exposes results to the somewhat arbitrary articulation of the road network topology into elements, especially in regard to the disaggregation introduced by modellers for the optimal representation of supply. To have full control over driver perception of this aspect it is possible to apply the proposed model to predefined sequences of arcs.

3.1.5. Summary of the awareness model

The proposed awareness model covers the most significant sources of information present in contemporary road networks. ICM can then be applied in the context of any combination of awareness sources reflecting actual or hypothetical ATIS scenarios. Table 1 demonstrates how the joint probability is sensitive to introducing additional sources and market penetration changes, further illustrated with numerical examples on Sioux-Falls network in Section 5.2.

Table 1
Joint awareness probability for multiple sources in various ATIS scenarios.

ATIS scenario	Awareness probability of a source:					Joint probability (Eq. (24))
	Observation	Radio	Online	VMS	Radio #2	
1	0.2	–	–	–	–	0.20
2	0.2	0.2	–	–	–	0.36
3	0.2	0.2	0.2	–	–	0.49
4	0.2	0.2	0.2	0.2	–	0.59
5	0.2	0.2	0.8	0.2	–	0.90
6	–	0.2	–	–	0.2	0.36

3.2. Rerouting model

The second step of ICM is aimed at representing the rerouting decision process of drivers who have already become aware about the unexpected event and (as argued in Section 3.1) figured out the new cost pattern.

The proposed rerouting model will be evaluated repeatedly within the DTA algorithm (for each node, instant, destination, and iteration). Because our requirement is the applicability of the method to large networks, the proposed model, apart from being methodologically sound and realistic, needs to be light in terms of computation resources. Instead, the existing models require relatively heavy calculation: correlation matrix (Gao et al., 2008), path searches (Pel et al., 2009), or agent within-day replanning (Dobler et al., 2012).

We propose a generic structure where the meaningful variables to express the utility of rerouting are provided at each decision point. Here we use the simple binomial logit model; it can be however replaced with more complex Random Utility Models. The model determines the discrete choice probability that aware drivers will reroute or not, subject to their current location, their final destination and the traffic conditions that they expect. More specifically, the rerouting model yields the share $\kappa_i^d(\tau)$ of aware drivers exiting from node $i \in N$ at time τ and directed towards destination $d \in Z$ who decide to change path.

The two main reasons for which the aware drivers will reroute are:

- the availability of a better alternative which differs substantially from the typical path, i.e. to seek for gains;
- the estimate of significantly higher costs on the typical path wrt those on the currently optimal route choice, i.e. to avoid losses.

The systematic utility of rerouting from node i at time τ is then defined as a function of:

- $\Delta p_i^d(\tau)$, *expected gain* – which reflects the change of user optimal paths in terms of arc conditional probabilities between rerouting (based on actual costs) and not rerouting (based on typical costs);
- $\Delta w_i^d(\tau)$, *avoided loss* – which reflects the performance difference in terms of average arc potentials between rerouting (based on actual paths) and not rerouting (based on typical paths).

Both of the above variables are derived from the expected costs to get to the destination. The optimal path to possibly reroute can differ substantially in terms of topology from the typical path, even in presence of modest cost differences; on the other hand, relevant cost differences usually induce major changes in the optimal path. So, not all combinations of low and high expected gains and avoided losses are possible.

3.2.1. Expected gain

The conditional probabilities of the forward star of the generic node reflect the expected minimal costs to reach the destination using the corresponding arcs, subject to the behavioural assumptions on route choice model. Arc conditional probabilities change if new optimal paths are found, while, clearly, no gain can be obtained by travelling on the typical path. The fact that the actual (optimal) arc conditional probabilities are different from the typical (obsolete) ones indicates that the driver can gain by rerouting as compared to the baseline, i.e. propagate along typical probabilities. We then propose to measure the expected gain of rerouting in terms of how much the actual arc probabilities differ from the typical one. Note however that rerouting may or may not lead to a significant cost reduction, but it constitutes the necessary precondition to gain that.

The *cosine similarity* is here used to compare the vector of forward star probabilities obtained from the typical costs and times, denoted $\hat{\mathbf{p}} = \text{RCM}(\hat{\mathbf{c}}, \hat{\mathbf{t}})$, with that obtained from actual costs and times, denoted $\mathbf{p} = \text{RCM}(\mathbf{c}, \mathbf{t})$:

$$\Delta p_i^d(\tau) = 1 - \frac{\sum_{a \in A_i^+} p_a^d(\tau) \cdot \hat{p}_a^d(\tau)}{\sqrt{\sum_{a \in A_i^+} p_a^d(\tau)^2} \cdot \sqrt{\sum_{a \in A_i^+} \hat{p}_a^d(\tau)^2}}. \quad (35)$$

In the above expression Δp increases for decreasing vectors similarity. Referring to the case of Fig. 8, as intended we have: $\Delta p = 0$, for parallel vectors (0,1) and (0,1), i.e. if arc conditional probabilities did not change; Δp increases as the

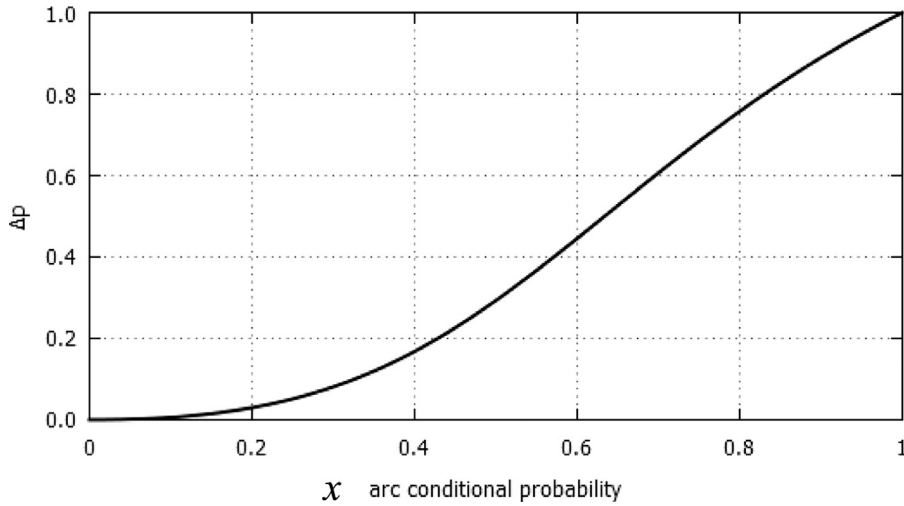


Fig. 8. Expected gain Δp computed for vectors $(0, 1)$ and $(x, 1 - x)$ as a function of x .

Table 2

Expected gain of rerouting Δp computed by cosine similarity for various situations of arc conditional probabilities at a generic node with three outgoing links (left, straight, right).

Situation	Arc conditional probabilities p_a^d						Δp (Eq. (35))
	Typical			Actual			
	Left	Straight	Right	Left	Straight	Right	
1	0	1	0	0	1	0	0.00
2	0	1	0	1	0	0	1.00
3	0	1	0	0.6	0	0.4	1.00
4	0	1	0	0	0.9	0.1	0.01
5	0	1	0	0	0.1	0.9	0.89
6	0	1	0	0.3	0.3	0.4	0.49
7	0.2	0.8	0	0	0.8	0.2	0.06
8	0.3	0.4	0.3	0.3	0.3	0.4	0.03

difference between the two vectors increases, reaching $\Delta p = 1$, for orthogonal vectors $(0,1)$ and $(1,0)$, i.e. when the arc conditional probabilities changed diametrically. Interestingly, the resulting non-linear shape implicitly provides a smoothed version of the indifference threshold for rerouting, which covers the hypothesis of [Mahmassani and Jayakrishnan \(1991\)](#) that drivers change path only when differences in time to get to destination become significant.

3.2.2. Avoided loss

To estimate the loss avoided by a rerouting driver we evaluate the relative difference in the two options of his *expected costs*. The latter is defined as the sum of the forward star arc potentials $w_a^d(\tau)$ weighted with the corresponding conditional probabilities.

If the driver does not reroute and will follow his trip according to the typical arc conditional probabilities, his expected cost will be:

$$\sum_{a \in A_i^+} \hat{p}_a^d(\tau) \cdot w_a^d(\tau). \quad (36)$$

Instead, if the driver does reroute according to the actual arc conditional probabilities, his expected cost will be:

$$\sum_{a \in A_i^+} p_a^d(\tau) \cdot w_a^d(\tau). \quad (37)$$

The avoided loss of rerouting is then given by:

$$\Delta w_i^d(\tau) = 1 - \frac{\sum_{a \in A_i^+} p_a^d(\tau) \cdot w_a^d(\tau)}{\sum_{a \in A_i^+} \hat{p}_a^d(\tau) \cdot w_a^d(\tau)}. \quad (38)$$

Table 3

Expected gain Δp and avoided loss Δw of rerouting computed for several traffic situations at a generic node with two outgoing links (left and straight).

Situation	w_a^d (actual costs)		Arc conditional probabilities p_a^d				Expected cost		$\Delta p(\text{Eq. (35)})$	$\Delta w(\text{Eq. (38)})$
	Left	Straight	Typical		Actual		Of not rerouting	Of rerouting		
			Left	Straight	Left	Straight				
1	4	3	0	1	0	1	3.0	3.0	0	0.00
2	2.9	3	0	1	1	0	3.0	2.9	1	0.03
3	4	30	0	1	1	0	30.0	4.0	1	0.87

Table 3 shows that Δw becomes significant only when costs on the typical path are higher wrt to the optimal route choice, while Δp is high as soon as new optimal paths are found.

3.2.3. Binomial rerouting logit model

The two above indicators can be synthesized into the rerouting utility of a discrete choice model. The rerouting utility for drivers being at node $i \in N$ at time τ while travelling towards destination $d \in Z$ is then a random variable with *systematic* component $v_i^d(\tau)$, which is a linear function of expected gain $\Delta p_i^d(\tau)$ and avoided loss $\Delta w_i^d(\tau)$ with given parameters β :

$$v_i^d(\tau) = \beta_0 + \beta_{\Delta p} \cdot \Delta p_i^d(\tau) + \beta_{\Delta w} \cdot \Delta w_i^d(\tau). \quad (39)$$

The proposed formulation of the systematic utility is here applied in a binomial logit model where the two alternatives are: do and do not reroute. The systematic utility of not rerouting is assumed zero, because that of rerouting is based on gains and losses, which already express a utility difference. The output of the binomial rerouting model is the probability $\kappa_i^d(\tau)$ that aware drivers directed to destination d will change path being at node i at time τ :

$$\kappa_i^d(\tau) = \frac{1}{1 + \text{Exp}(-v_i^d(\tau))}. \quad (40)$$

Model calibration is out of scope of the paper; for more details on the estimation problem we refer to Kucharski and Gentile (2015). Here we just discuss how to obtain a desired rerouting behaviour by tuning the parameters β of the systematic utility. In general, small values of betas lead to casual rerouting, barely sensitive to changes in the Δp and Δw , while large betas will lead to high sensitivity. Since Δp and Δw are normalized, the share between risk-prone aggressive drivers seeking for gains and risk-averse conservative drivers avoiding losses can be directly controlled via the ratio of $\beta_{\Delta p}$ and $\beta_{\Delta w}$, while β_0 controls the general willingness to reroute.

Parameterization may be tested for boundary cases:

- for no rerouting utility (Δp and Δw are zero and the probability is a function of β_0 only), κ shall be infinitesimal;
- for maximal utility (Δp and Δw reaching 1), κ should reflect a high rerouting share, including the fact that some aware drivers will never reroute, regardless the utility.

The proposed utility formulation and the hypotheses of non-linear rerouting behaviour (e.g. indifference band of Mahmassani and Jayakrishnan, 1991; or Prospect Theory behaviour of Gao et al., 2010) shall be verified while estimating the model with the field-data.

In the numerical examples presented later in this paper, we found a suitable sensitivity of the rerouting model using parameters: $\beta_{\Delta p} = 2.5$, $\beta_{\Delta w} = 2.5$ and $\beta_0 = -5$. With this configuration the rerouting probability is close to zero when there is no rerouting utility and reaches 50% when Δp and Δw equal 1, as shown in Table 4.

Table 4

Rerouting probabilities computed for various potential gains and expected losses.

Δp	Δw	v	κ
0.00	0.00	−5.000	00.67%
0.10	0.01	−4.725	00.88%
0.30	0.01	−4.225	01.44%
0.50	0.01	−3.725	02.35%
0.70	0.01	−3.225	03.82%
0.90	0.01	−2.725	06.15%
1.00	0.01	−2.475	07.76%
0.90	0.30	−2.000	11.92%
0.90	0.50	−1.500	18.24%
0.90	0.70	−1.000	26.89%
1.00	1.00	−0.000	50.00%

To enhance numerical stability and due to issues related with the long tails of logit model and its sequential application, which are discussed later in the paper, we introduced a threshold for rerouting probabilities: when κ is below 0.02 we assume it to be zero.

3.3. Rerouting path

In this step of ICM we reproduce how drivers choose a new path from the node where they decided to reroute to their destination. Generally, drivers update their knowledge on the network performance pattern with the information they have received and using their own prediction of the travel times resulting from the event.

The cognitive and decision process of predicting the future travel costs and reacting to them is not only complex, but also nondeterministic, rarely rational, strongly dependent to the type of information and thus still remains an opened research question (see, for example, Ramos (2015) and Selten et al. (2007) for more insights on how informed travellers predict costs).

To formalize the model, we argue that drivers make rerouting decisions based on accurate information. We follow the classic law of information provision by Bifulco et al. (2016) stating that the forecasted travel costs shall be in line with the actually experienced ones. Yet, since complying to the information alters the costs that user was informed about, the fixed-point problem arises. At the convergence of this fixed-point problem, rerouting drivers utilize the information on forecasted evolution of the system including anticipated reaction of drivers to that information. Accordingly, the rerouting drivers will experience the rerouting costs they were informed about. We propose the ICM model, where results are obtained after solving a fixed-point problem between information and drivers reaction to that information (we illustrate the convergence and intermediate results in the Sioux-Falls example in Section 5.2).

Thanks to such approach, ICM model can be formalized with assumption that rerouting drivers use the arc conditional probabilities, $\mathbf{p} = RCM(\mathbf{c}, \mathbf{t})$, computed with the RCM based on the actual costs and times. Clearly, at each iteration the costs may change since the rerouting flows will yield different costs. Likewise, the arc probabilities used for rerouting will change, until finally reaching convergence of this particular type of equilibrium (not fully in line with Wardrop principles though), where not all drivers are perfectly informed and comply with a rational behaviour.

While the above assumptions may seem unrealistic and doubtful, they allow to keep the model deterministic and formal. Assuming that the rerouting drivers do not anticipate others rerouting decisions at all would be even more unrealistic. Assuming partial (incomplete) anticipation would be arbitrary and intractable. Assuming perfect anticipation allows to obtain a formal model without heuristics, which allows for scenario comparison and benchmarking.

Notably, proposed model does necessarily assume exceptional drivers predictive capabilities, since the information may be externally provided. Moreover, we believe that for rerouting phenomena, the effect of information spread and compliance is much more significant than the path choice process when the rerouting decision is made. Critical is when and where the driver receives the information rather than the assumptions on rerouting path choice behaviour, which do not change the essence of rerouting picture resulting from ICM.

Also, the route choice model selection is also of high importance. Particular attention is to be paid for deterministic models, whose choice map is one-to-many. In this case, if all-or nothing assignment is considered in the demand model, small changes of costs will yield substantial changes in arc conditional probabilities. This can be avoided by considering at every iteration of the equilibrium method the probabilities provided by gradient projection algorithms, as in Gentile (2016), which depend smoothly on arc performances. In the numerical experiments we mainly utilize deterministic route-choice model not to make confusion between the stochasticity of the route choice model and that of the information and comply process. Yet if the non-deterministic driver behaviour needs to be reproduced, stochastic model can be used as shown in one of numerical tests. Importantly, the rerouting route-choice model can be differently parameterized than the typical RCM to capture more stochastic nature of rerouting choices, different utility formulation etc.

4. Combining the ICM into the DTA

4.1. Demand model based on ICM

In this section the ICM formally introduced in the previous section is coupled with the DTA structure introduced in Section 2. The ICM was conceived to be methodologically sound and to realistically represent the actual behaviour, but also to be seamlessly and efficiently integrated with the DTA model, so as to reproduce rerouting phenomena on real-size network in a suitable computation time.

To integrate ICM in the macroscopic DTA model the propagated flow $\mathbf{q}$ is partitioned into the three conditions:

- unaware $\hat{\mathbf{q}}$,
- aware $\bar{\mathbf{q}}$ and
- rerouted $\tilde{\mathbf{q}}$.

These flows are propagated simultaneously with two possible subsequent transitions:

- awareness probability ι , when unaware drivers become aware along arcs, and

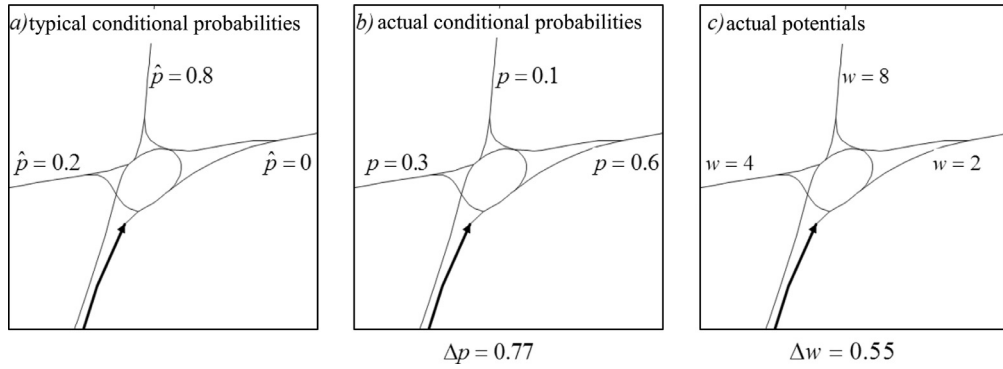


Fig. 9. Example of rerouting systematic utility at a generic node where arc conditional probabilities changed significantly due to event. The driver approaches the intersection from the bottom and makes his rerouting choice based on the following variables of the forward star arcs: a) typical conditional probabilities, b) actual conditional probabilities (different from the expected ones due to the event downstream), c) actual potentials.

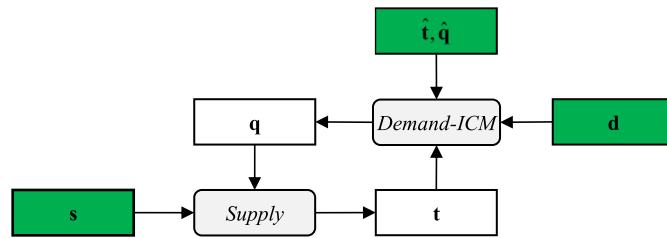


Fig. 10. Simplified schema of the DTA model with Information Comply Model.

- rerouting probability κ , when aware drivers reroute at nodes.

Unaware and aware flows are propagated accordingly with the typical arc conditional probabilities $\hat{\mathbf{p}}$, which result from the dynamic user equilibrium of the day-type, while rerouted flows are propagated accordingly with the actual probabilities $\mathbf{p}$, which result from the new assignment including the information comply process of the simulated event. All flow conditions (also unaware) are propagated accordingly with the actual travel times.

To integrate the above concept into DTA, the flow propagation model is redefined as follows. First, the typical conditional probabilities can be easily obtained from the typical destination flows, by considering the share of arc flows exiting from the tail node:

$$\hat{p}_a^d(\tau) = \frac{\hat{q}_a^d(\tau)}{\hat{q}_{i=N_a^-}^d(\tau)} = \frac{\hat{q}_a^d(\tau)}{\sum_{b \in A_i^+} \hat{q}_b^d(\tau)}. \quad (41)$$

The demand starts at origins as unaware flow. This is progressively reduced along the arcs by the awareness probability and propagated toward the destination accordingly with the typical conditional probabilities:

$$\hat{q}_i^d(\tau) = d_{id}(\tau) + \sum_{a \in A_i^-} \hat{q}_a^d(\theta_a^{-1}(\tau)) \cdot (1 - \iota_a(\theta_a^{-1}(\tau))) \cdot \frac{\partial \theta_a^{-1}(\tau)}{\partial \tau}, \quad (42)$$

$$\hat{q}_a^d(\tau) = \hat{q}_{N_a^-}^d(\tau) \cdot \hat{p}_a^d(\tau). \quad (43)$$

Recall that all the demand starts at origins at time 0 and exits at the desired departure time from a fictitious arc which allows to correctly initialize the aware flows to take into account the information possibly obtained before starting the trip.

The aware flow is first increased with the share of unaware flow given by the awareness probability, then reduced by the rerouting probability, and finally propagated toward the destination still in line with the typical conditional probabilities:

$$\tilde{q}_i^d(\tau) = (1 - \kappa_i^d(\tau)) \cdot \sum_{a \in A_i^-} (\tilde{q}_a^d(\theta_a^{-1}(\tau)) + \hat{q}_a^d(\theta_a^{-1}(\tau)) \cdot \iota_a(\theta_a^{-1}(\tau))) \cdot \frac{\partial \theta_a^{-1}(\tau)}{\partial \tau}, \quad (44)$$

$$\tilde{q}_a^d(\tau) = \tilde{q}_{N_a^-}^d(\tau) \cdot \hat{p}_a^d(\tau). \quad (45)$$

Finally, the rerouted flow is increased with the share of both already and newly rerouted flow given by the rerouting probability, and propagated toward the destination, now in line with the actual conditional probabilities:

$$\tilde{q}_i^d(\tau) = \sum_{a \in A_i^-} (\tilde{q}_a^d(\theta_a^{-1}(\tau)) + \kappa_i^d(\tau) \cdot (\tilde{q}_a^d(\theta_a^{-1}(\tau)) + \hat{q}_a^d(\theta_a^{-1}(\tau)) \cdot \iota_a(\theta_a^{-1}(\tau)))) \cdot \frac{\partial \theta_a^{-1}(\tau)}{\partial \tau}, \quad (46)$$

$$\tilde{q}_a^d(\tau) = \tilde{q}_{N_a^-}^d(\tau) \cdot p_a^d(\tau). \quad (47)$$

This can be seen as Markov chain where transition probabilities of different kinds are involved, including the share of flow entering on an arc during a given interval and exiting it at a later interval, i.e. the propagation map, which will substitute the time derivatives in the algorithm implementation.

The new version of the FPM with ICM expresses in compact form the Eqs. (41)–(47):

$$\mathbf{q} = \text{FPM-ICM}(\mathbf{p}, \mathbf{t}, \iota, \kappa; \hat{\mathbf{q}}; \mathbf{d}). \quad (48)$$

Based on the formulation presented in Section 3.1 we can obtain the awareness probability of the generic arc $a \in A$ at time $t \in T$ through the Event Awareness Model (EAM) as a function of the actual and typical travel time and of the total network time relative increment:

$$\iota_{at} = \text{EAM}(\tau_t, t_{at}, \hat{t}_{at}, \Delta t_t). \quad (49)$$

Based on the formulation presented in Section 3.2 we can obtain the rerouting probability to destination $d \in Z$ from the generic node $i \in N$ at time $t \in T$ through the Event Routing Model (ERM) as a function of the actual and typical conditional probabilities and of the actual potentials of each arc in the node forward star:

$$\kappa_{it}^d = \text{ERM}(\hat{p}_{at}^d, p_{at}^d, w_{at}^d : a \in A_i^+). \quad (50)$$

Note that, while rerouting probabilities are destination specific and shall then be computed together with the conditional probabilities, the awareness probabilities are not and can then be computed at once for all destinations.

Finally, the new version of the demand model with ICM is the composition of ACM, EAM, APM, LRM, ERM, FPM, where the profile vector version of the EAM includes the computation of the relative time increments (31) and the ERM includes the computation of the typical conditional probabilities from the destination flows (41):

$$\text{Demand-ICM}(\mathbf{t}; \hat{\mathbf{t}}, \hat{\mathbf{q}}; \mathbf{d}) = \text{FPM}(\mathbf{p} = \text{LRM}(\mathbf{w} = \text{APM}(\text{ACM}(\mathbf{t}), \mathbf{t})), \mathbf{t}, \text{EAM}(\mathbf{t}; \hat{\mathbf{t}}, \hat{\mathbf{q}}), \text{ERM}(\mathbf{p}, \mathbf{w}; \hat{\mathbf{q}}); \hat{\mathbf{q}}; \mathbf{d}). \quad (51)$$

The following Algorithm 2 computes the demand model with ICM, taking a fixed input the typical traffic pattern $(\hat{\mathbf{t}}, \hat{\mathbf{q}})$, while the actual travel times are provided by the supply model and will vary across iterations.

Note that there is no need to memorize the different kinds of arc flows (unaware, aware, rerouted) between iterations as only their aggregated value is relevant in the supply model and for the convergence. In the practical implementation of the algorithm, only two vectors of variables by destination are saved in memory: the actual arc flows and the typical arc flows, from which it is immediate to obtain the conditional probabilities.

4.2. DTA with Information Comply Model

The combination of the two sub-models Demand-ICM (51) and Supply (1) yields a fixed point problem which formalizes the Dynamic Traffic Assignment with Information Comply Model:

$$(\mathbf{t}, \mathbf{q}) = \text{DTA-ICM}(\mathbf{s}, \mathbf{d}; \hat{\mathbf{t}}, \hat{\mathbf{q}}). \quad (52)$$

This can still be seen as a kind of user equilibrium (not fully in line with Wardrop principles though), i.e. a traffic condition where no driver can improve his perceived path cost, including both awareness of the event and reluctance to change.

DTA-ICM provides usual DTA output (travel times and destination flows). The input are supply and demand including any direct event effect, as well as its position in time and space, the broadcast times, the time when the information enters then monitoring system, etc. As an additional input, the typical traffic conditions and flows are needed, which can be obtained by solving the DTA model for the typical supply $\hat{\mathbf{s}}$ and demand $\hat{\mathbf{d}}$:

$$(\hat{\mathbf{t}}, \hat{\mathbf{q}}) = \text{DTA}(\hat{\mathbf{s}}, \hat{\mathbf{d}}). \quad (53)$$

4.3. Convergence of MSA and intermediate results

The DTA-ICM is formulated as fixed point problem, which can be solved by repeatedly applying the following contraction formula:

$$\mathbf{q}^n = (1 - \alpha_n) \cdot \mathbf{q}^{n-1} + \alpha_n \cdot \text{Demand-ICM}(\text{Supply}(\mathbf{q}^{n-1}; \mathbf{s}); \hat{\mathbf{t}}, \hat{\mathbf{q}}; \mathbf{d}). \quad (54)$$

Algorithm 2 Demand model with Information Comply Model.

```

function Demand-ICM(t, e, m;  $\hat{\mathbf{t}}$ ,  $\hat{\mathbf{q}}$ ; d)
     $t_{tot} \leftarrow 0$ ,  $\hat{t}_{tot} \leftarrow 0$                                 * reset total times
    for  $t = 0$  to  $\mu-1$                                        * for each interval in chronology
         $t_{tot} \leftarrow t_{tot} + t_{at} \cdot \hat{q}_{at}$                 * compute the actual potential total time
         $\hat{t}_{tot} \leftarrow \hat{t}_{tot} + t_{at} \cdot \hat{q}_{at}$                 * compute the typical potential total time
         $\Delta t_t \leftarrow \hat{t}_{tot} / t_{tot} - 1$                   * compute the total time relative increment
    next  $t$ 
     $t_{at} \leftarrow EAM(\tau_t, t_{at}, \hat{t}_{at}, \Delta t_t) \quad \forall a \in A, \forall t \in T$  * event awareness model
     $c_{at} \leftarrow ACM(t_{at}) \quad \forall a \in A, \forall t \in T$           * compute travel costs
    for each  $d \in Z$                                            * for each destination
        

---


        first node visit in chronological order
         $w_{AT}^d \leftarrow APM(\mathbf{c}, \mathbf{t}, \mathbf{e}, d)$                 * compute arc potentials as in Algorithm 1
        

---


        second node visit in chronological order
         $\hat{\mathbf{q}}_{NT}^d \leftarrow \mathbf{0}_{NT}$ ,  $\bar{\mathbf{q}}_{NT}^d \leftarrow \mathbf{0}_{NT}$ ,  $\tilde{\mathbf{q}}_{NT}^d \leftarrow \mathbf{0}_{NT}$  * reset all node flows
        for  $t = 0$  to  $\mu-1$                                        * for each interval in chronology
            for each  $i \in N$                                        * process each node in any order
                 $\mathbf{p}_{A_i^+}^d \leftarrow p(w_{bt}^d : b \in A_i^+)$         * arc probabilities for the given choice model
                 $\kappa_{it}^d \leftarrow ERM(\hat{p}_{at}^d, p_{at}^d, w_{at}^d : a \in A_i^+)$  * event rerouting model
                if  $i \in Z$  then  $\tilde{q}_{it}^d \leftarrow \tilde{q}_{it}^d + d_{idt}$     * load the od demand on the node if an origin
                for each  $a \in A_i^+$                                 * for each arc of the forward star
                     $\hat{q}_{at}^d \leftarrow \hat{q}_{it}^d \cdot \hat{p}_{at}^d$             * compute the unaware flow
                     $\bar{q}_{at}^d \leftarrow \bar{q}_{it}^d \cdot \hat{p}_{at}^d$             * compute the aware flow
                     $\tilde{q}_{at}^d \leftarrow \tilde{q}_{it}^d \cdot p_{at}^d$             * compute the rerouted flow
                     $q_{at}^d \leftarrow \hat{q}_{at}^d + \bar{q}_{at}^d + \tilde{q}_{at}^d$         * aggregate the total flow (for the supply model)
                 $j \leftarrow N_a^+$                                 * consider the head node
                for  $e = e_{at}$  to  $e_{at+1}$                         * for each exiting interval, compute
                     $\hat{q}_{je}^d \leftarrow \hat{q}_{je}^d + \hat{q}_{at}^d \cdot (1 - t_{at}) \cdot m_{ate}$  * the unaware node flow
                     $\bar{q}_{je}^d \leftarrow \bar{q}_{je}^d + (\bar{q}_{at}^d + \hat{q}_{at}^d \cdot t_{at}) \cdot m_{ate} \cdot (1 - \kappa_{te}^d)$  * the aware node flow
                     $\tilde{q}_{je}^d \leftarrow \tilde{q}_{je}^d + \tilde{q}_{at}^d \cdot m_{ate} + (\bar{q}_{at}^d + \hat{q}_{at}^d \cdot t_{at}) \cdot m_{ate} \cdot \kappa_{te}^d$  * the rerouted node flow
                next  $e$ 
            next  $a$ 
        next  $i$ 
    next  $\tau$ 
    

---


    next  $d$ 
    return  $\mathbf{q}$                                                 * return destination flows
end function

```

The solution is obtained when a fixed-point is reached, while rerouting is adjusted iteration by iteration. In the intermediate iterations the travel costs perceived by rerouted drivers while making their path choices are not optimal. This can be reflected by the following relative gap where only the already rerouting drives are considered:

$$\rho_n = \sum_{a \in N, d \in Z} \int_0^\tau \tilde{q}_{Na}^d(\tau) (w_a^d(\tau) - w_{Na}^d(\tau)) \cdot d\tau, \quad (55)$$

The gap ρ_n tends to zero as the DTA-ICM converges to a fixed-point along with rerouting drivers reaching perfect prediction of actual travel costs. Intermediate solutions, when ρ_n is still above zero, represent drivers affected by imperfect travel costs prediction when making rerouting decision. Although these intermediate results are not obtained through a proper model but rather depend on the algorithm convergence, they can be considered as possible outcome of a simulation process and have then some explanatory value.

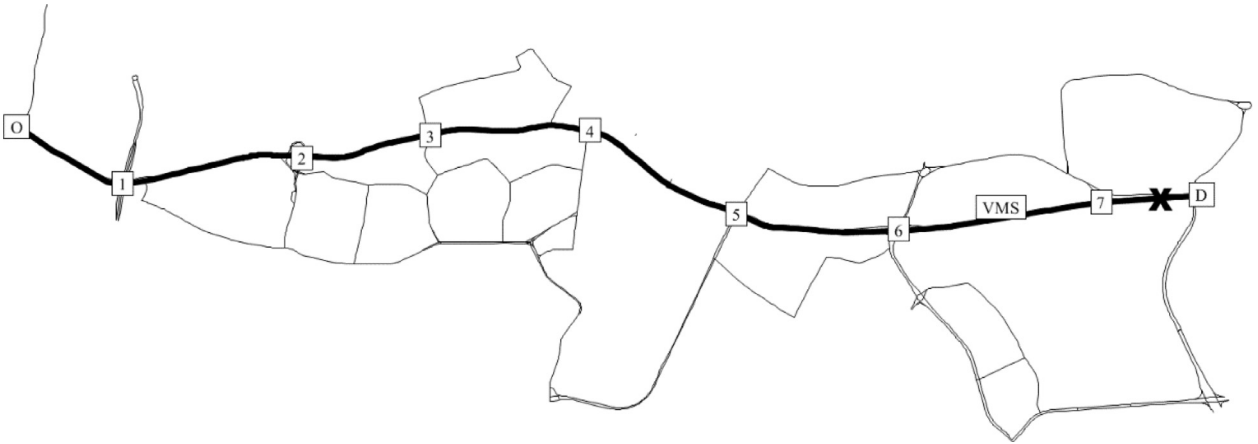
4.4. Implementation

We implemented the DTA-ICM in the framework of the TRE software, which is the simulation engine of Optima (PTV SIS-TeMA, 2016). We tested its performance on the Sioux-Falls network and not only obtained meaningful results (reported

Table 5

Computation times of the DTA-ICM algorithm for the Sioux-Falls network.

Network				Classic DTA		DTA-ICM							
Arcs	Zones	Time period	Time intervals	Comp. time	Demand output size	Comp. time	Of which:						
						Supply	Shortest path	Load typical demand	RCM	FPM with awareness model	Rerouting model	Other	
1306	24	4[h]	12[s]	8[s]	350[MB]	18[s]	6%	11%	0.04%	7%	60%	0.65%	15%

**Fig. 11.** Corridor example. A path (bold) connecting origin O with destination D, passing through 7 decision points and one VMS sign just before the event marked with X.

in Section 5.2), but also acceptable computation times. The DTA computation times after implementing ICM (reported in Table 5) grown from 8 to 18 s per iteration, and the increase was mainly due to recurrent Rayleigh PDF evaluation (Eq. (29)). This is a significant contribution, since this is (up to our knowledge) the first DTA method to reproduce rerouting behaviour in acceptable time.

5. Numerical experiments

We present here some experiments of the proposed Information Comply Model on test networks. We start with a generic corridor example along which we propagate 100 vehicles in three different scenarios exposing rerouting phenomena picture of ICM. Then we simulate the event on the Sioux-Falls network and report temporal profile of flows in five different ATIS. Finally, we report the algorithm convergence and stability in the case of limited capacity.

5.1. Corridor detour

We illustrate ICM with a corridor in the north of Kraków, Poland (Fig. 11) consisting of 7 nodes at which the rerouting decision can be made. Close to the destination, at 7^{10} , there is an event (X) blocking the arc, so that the typical route is no longer optimal and aware drivers start seeking for alternatives. We simulate the flow of 100 vehicles departing from the origin o at 7^{00} and propagating towards the destination d . Table 6 shows at consecutive nodes: flows in three conditions, awareness probability of respective sources, rerouting probabilities and potential gains and losses of rerouting. Rerouting utility parameters are $\beta_{\Delta p} = 2.5$, $\beta_{\Delta w} = 2.5$ and $\beta_0 = -5$ and the lower bound for κ of 5% was used (the rerouting probability below 5% is assumed to be zero). We present results for the cases of: a) multiple sources of information, b) no information sources and c) information available at the origin. To keep the examples demonstrative, market penetrations of information sources were intentionally magnified, so that almost everybody becomes aware.

Table 6 presents the results for the case when the information is broadcasted through the radio, available on-line and shown on the VMS. ICM modelled 95% of the drivers becoming aware and 36% actually rerouting. At the 1st node there is no information available, so the whole flow remains unaware. During propagation from 1st to 2nd node (7^{00} to 7^{10}), 3% of the drivers became aware of the event by browsing on-line sources, yet none of them decides to reroute as the utility remains null. At 7^{17} the information about the event is broadcasted on the radio, with increment of aware drivers during that broadcast $\iota^{NEWS} = 30\%$. The drivers propagating from 3rd to 4th node at that time become aware. At 7^{20} the event starts impacting the whole network and the Δt reaches 4% at 8^{20} , triggering virality of on-line information which now spreads

Table 6

ICM results at 7 consecutive nodes traversed during propagation along a corridor. Information broadcasted at the radio, on-line and at the VMS.

Node		Node flow			Awareness model							Rerouting model		
<i>i</i>	τ	$\hat{q}$	$\bar{q}$	$\bar{q}^1$	ι	$\iota_a^{NEWS}(\tau)$	$\iota_a^{VMS}(\tau)$	$\iota_a^{ONLINE}(\tau)$	$\Delta t^{tot}(\tau)$	$\iota_a^{OBS}(\tau)$	Δt_a ²	κ	Δp	Δw
1	7:00	100	0	0	0%	–	–	0%	–	–	–	–	0.00	0.00
2	7:10	97	3	0	3%	–	–	3%	–	–	–	–	0.01	0.00
3	7:15	93	6	0	3%	–	–	3%	–	–	–	–	0.01	0.00
4	7:20	63	37	0	33%	30%	–	4%	1%	–	–	–	0.00	0.00
5	7:25	60	37	2	4%	–	–	4%	1%	–	–	6%	0.61	0.28
6	7:40	54	36	8	10%	–	–	10%	2%	–	–	18%	0.76	0.64
7	8:20	5	48	37	91%	–	80%	6%	4%	50%	500	43%	1.00	0.89

¹ we do not propagate the rerouted flows since they may reroute outside of the corridor, thus $\bar{q}$ in the table is only the increment($\bar{q} \cdot \kappa$).

² time increase in Eq. (34)

Table 7

ICM results at 7 consecutive nodes traversed during propagation along a corridor. No information about the event is available; drivers can only observe the atypical delay.

Node		Node flow			Awareness model							Rerouting model		
<i>i</i>	τ	$\hat{q}$	$\bar{q}$	$\bar{q}$	ι	$\iota_a^{NEWS}(\tau)$	$\iota_a^{VMS}(\tau)$	$\iota_a^{ONLINE}(\tau)$	$\Delta t^{tot}(\tau)$	$\iota_a^{OBS}(\tau)$	Δt_a	κ	Δp	Δw
1	7:00	100	0	0	–	–	–	–	–	–	–	–	–	–
2	7:10	100	0	0	–	–	–	–	–	–	–	–	–	–
3	7:15	100	0	0	–	–	–	–	–	–	–	–	–	–
4	7:20	100	0	0	–	–	–	–	1%	–	–	–	–	–
5	7:25	100	0	0	–	–	–	–	2%	–	–	19%	0.92	0.50
6	7:40	70	16	14	30%	–	–	–	3%	30%	300	46%	0.98	0.95
7	8:20	42	22	22	40%	–	–	–	5%	40%	400	50%	1.00	1.00

Table 8

ICM results at 7 consecutive nodes traversed during propagation along a corridor. The event is known by 30% of drivers at the departure time 8:00. Information broadcasted at the radio, on-line and at the VMS.

Node		Node flow			Awareness model							Rerouting model		
<i>i</i>	τ	$\hat{q}$	$\bar{q}$	$\bar{q}$	ι	$\iota_a^{NEWS}(\tau)$	$\iota_a^{VMS}(\tau)$	$\iota_a^{ONLINE}(\tau)$	$\Delta t^{tot}(\tau)$	$\iota_a^{OBS}(\tau)$	Δt_a	κ	Δp	Δw
1	8:00	69	31	0	31%	–	–	31%	4%	–	–	–	0.00	0.00
2	8:10	69	31	0	–	–	–	–	4%	–	–	–	0.01	0.01
3	8:15	69	31	0	–	–	–	–	4%	–	–	–	0.01	0.00
4	8:20	48	51	0	30%	30%	–	–	4%	–	–	–	0.00	0.00
5	8:25	48	48	3	–	–	–	–	4%	–	–	6%	0.61	0.32
6	8:40	48	39	8	–	–	–	–	4%	–	–	17%	0.76	0.62
7	9:20	5	54	30	91%	–	80%	–	4%	50%	500	35%	1.00	0.75

faster and reaches more drivers. 80% of the drivers notice the VMS between node 6th and 7th. Since the event caused a delay at that arc (spillover), drivers become aware when experiencing it (educated guessing). Awareness for multiple sources reaches 91% at 7th node.

The rerouted flow remains null up to 5th node, although the 40% of the flow is already aware up to that time. Due to low utility in the rerouting model, drivers do not change path prior the 5th node, since there are no reasonable alternatives at nodes 1–4 and the actual demand pattern is still in-line with the typical. From 5th node on, where the alternative paths are available, the rerouting probability becomes positive. Δw is highest at the last node, where the last opportunity to avoid the event is available.

Table 7 presents the case when no information sources are available. Now the only source of awareness is the experienced delay Δt_a . Drivers remain unaware until they observe the consequences (spillover) which propagates backward to node 6 (since less people got aware and rerouted). Due to the lack of awareness, no one reroutes at the 5th node even though the κ is reasonably high. The rerouting takes place when the experienced drivers guess about the event from the delay (30% and 40% respectively), which leads to changing path at node 6 and 7. Yet, still 42% of the flow remains unaware and drives into the event.

Table 8 shows how ICM handles pre-trip rerouting by modelling the same corridor an hour later, when information is available on-line prior to departure. In this case 31% of drivers is aware already at the origin yet travel accordingly to the typical pattern until there is enough utility in rerouting. Since the rerouting utility did not change compared to the first example (Table 6), the change of path takes place from 5th node.

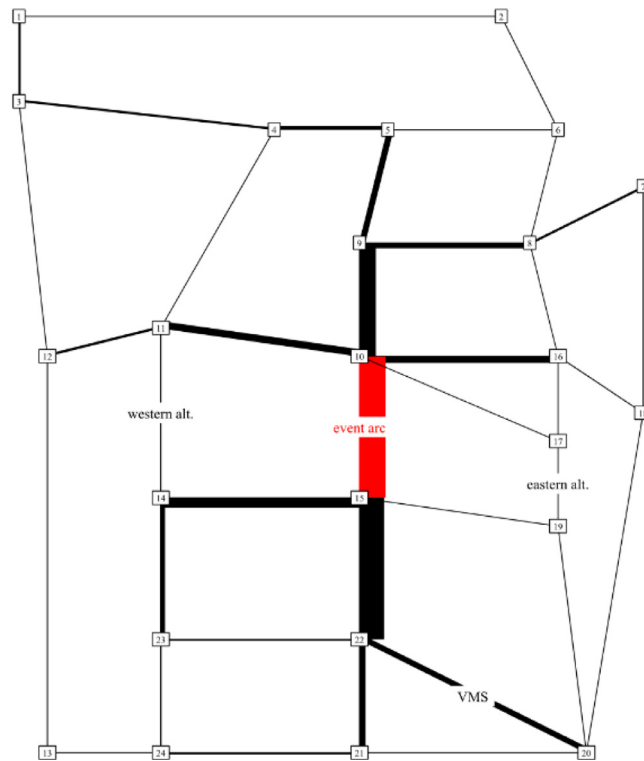


Fig. 12. Unexpected event in the Sioux-Falls network (red), vehicle flows directly affected the event marked with black bars. Two alternative arcs marked on Eastern and West side of the event. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 9
Rerouting scenarios at Sioux-Falls network.

Scenario	Description	Total rerouting flow	Queue length an hour past the event
No information	Drivers are unaware of the event	0	410
Online information	Broadcasted online 12 min past the event with spreading pace $\sigma = 1.5$ h	129	281
VMS	Placed upstream of event, visible to c.a. one third of flow, turned on 22 min past the event	30	380
Radio news	Radio news broadcasted 29 min past the event	116	294
Mixed deterministic RCM	All of the above simultaneously	156	254
Mixed stochastic RCM	All of the above simultaneously yet drivers reroute according to stochastic RCM	156	254

5.2. Rerouting in the Sioux-Falls network

We simulate the Sioux-Falls network where the event takes place at the central NS corridor northbound (Fig. 12). We simulate demand for 2 hours with the event taking place after an hour of simulation and limiting the arc capacity to 60 veh/h from this time forth. We conduct an experiment schema to illustrate DTA-ICM results. The demand is intentionally low (total 10 000 trips) to obtain uncongested cost pattern and draw attention to rerouting phenomena rather than to congestion. Like in the previous example, rerouting utility parameters are $\beta_{\Delta p} = 2.5$, $\beta_{\Delta w} = 2.5$ and $\beta_0 = -5$ and the lower bound for κ of 5% is used. ICM-DTA converges to ρ around 10^{-4} in 50 iterations which is enough to stabilize route choice decision, awareness processes, rerouting decisions as well as rerouting route choice decisions. To obtain clear picture of rerouting behaviour we use deterministic route-choice model (Eq. (11)) except of the last scenario where we apply the probabilistic model to illustrate the difference between the two.

The vehicles inflowing into the event arc build the queue and cause delays significant enough to make alternative routes attractive. The queue, however, does not reach the arc tail so that flows do not spillback. Aware drivers can avoid the event by rerouting to one of four alternative arcs, two of which are most attractive: western and eastern one.

We report temporal profiles of inflows into the event arc (Fig. 13) and the two alternative arcs (Figs. 14 and 15) in six different scenarios described in (Table 9). In no information is provided the steady flow of ca. 480 vehicles inflow to the event arc, when the arc becomes blocked, the queue grows to 410 vehicles an hour past the event. If no information is

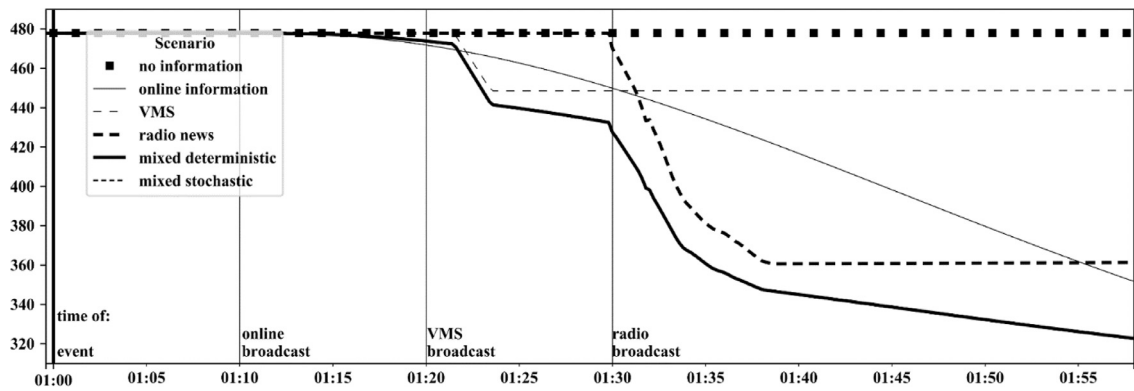


Fig. 13. Temporal profile of vehicle inflow (veh/h) into the event arc in simulated scenarios.

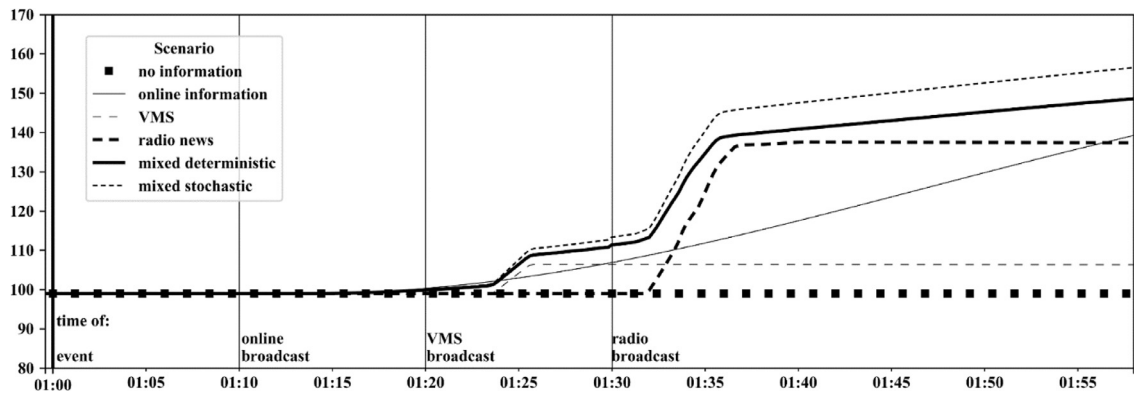


Fig. 14. Temporal profile of vehicle inflow into the western alternative arc in simulated scenarios.

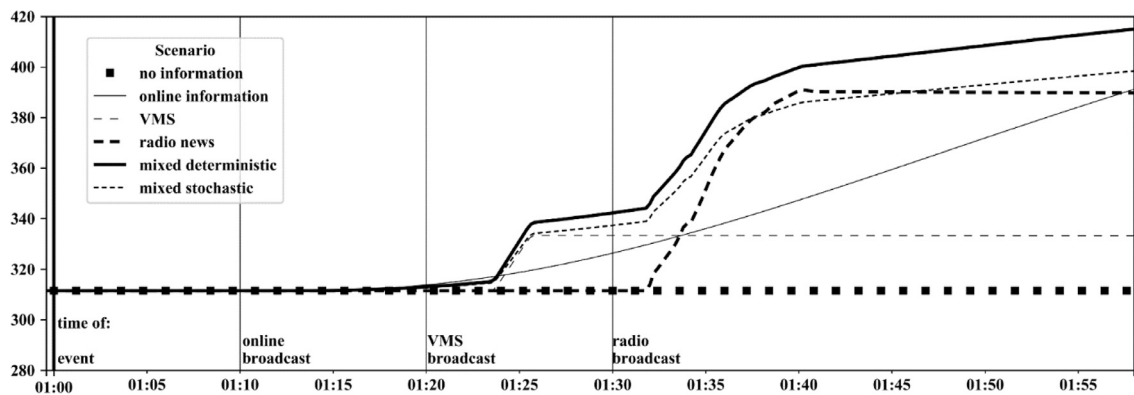


Fig. 15. Temporal profile of vehicle inflow into the eastern alternative arc in simulated scenarios.

provided the inflows at alternative arc also remain flat, since no one reroutes. If the information starts being spread online, it steadily reaches drivers who start rerouting, visible as the inflow decrease at the event arc and increase at alternative arcs, with total 129 rerouted drivers an hour past the event. This number as well as the shape of the curve can be controlled with the information source market penetration rate and rerouting model parameters to match the actual behaviour. The VMS is placed at one of the upstream arcs (Fig. 12), thus even if it is turned on, it is visible only by part of the drivers. The reaction is stable and around 30 aware drivers decides to reroute, mainly to the eastern alternative where the flow increases by more the 20 vehicles around 25 min past the event. The radio news of high penetration are broadcasted 30 min past the event. Drivers become aware whenever they are and instantly can make their rerouting decisions, resulting in rerouting 116 drivers. Therefore the decrease in inflows is smoother since drivers reroute from more decision points as compared to VMS case. The mixed case where all of the above leads into complex rerouting picture. In the first part no one is aware and the inflow to the event arc is steady. Then it decreases first as the information spreads online, then as the VMS is turned on and finally after the radio broadcast. In total 156 drivers reroutes and the queue drops from 410 to 254 vehicles. The inflow

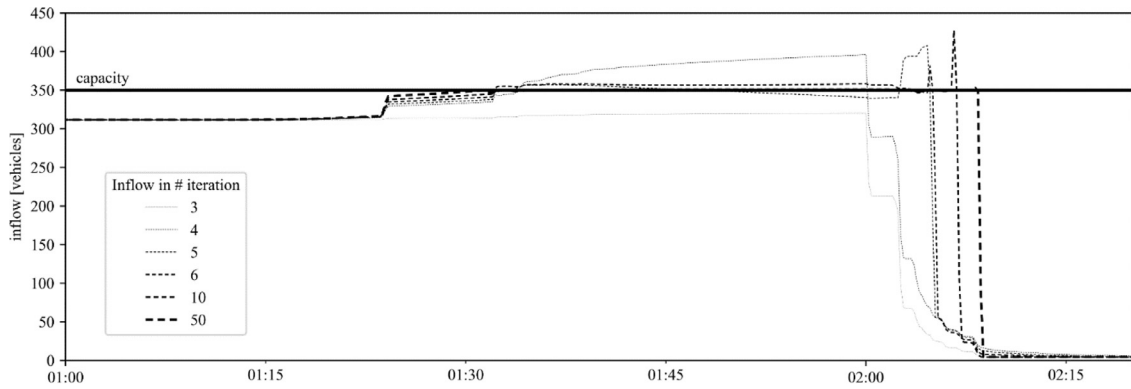


Fig. 16. Inflows into the eastern alternative arc of limited capacity in respective iterations.

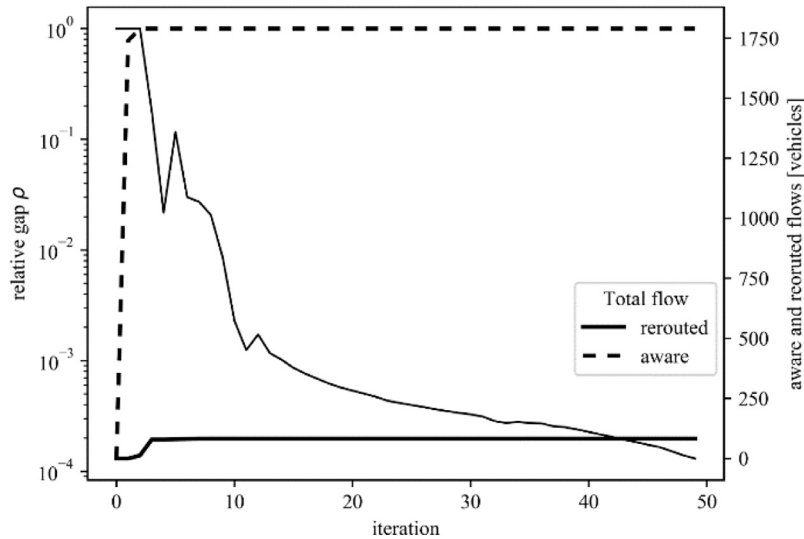


Fig. 17. Relative gap of rerouting decisions, rerouting and aware flows stability across iterations.

profile of the event arc does not change when we replace deterministic RCM with the probabilistic one, yet the rerouting flows are differently distributed to alternative arcs. The eastern alternative is now less popular to reroute since drivers act less optimally and are distributed among all feasible alternatives, including less efficient western one, where flows are now higher.

To illustrate algorithm convergence to the fixed-point we limited the capacity of eastern alternative to 350 veh/h, i.e. higher than typical flows (315 veh/h), but lower than flows in the mixed rerouting scenario (410 veh/h). We run 50 iterations of mixed scenario and report inflows in the 3rd, 4th, 5th, 6th, 10th and 50th iteration (Fig. 17). In the first iterations inflows exceed the capacity, i.e. rerouting drivers select path at which the queue builds up due to their decisions. Consequently their expected rerouting costs (both at the level of estimating rerouting utility and while making rerouting route-choice decision) differ from the actual ones that they will encounter. This however becomes consistent as the algorithm converges throughout the iterations. In 10th and 50th iterations the inflows are bounded by the capacity and drivers managed to optimize their rerouting decisions. ICM-DTA allows to model both perfect predictive information and optimal drivers decision (which is the formal model), as well as intermediate incomplete information obtained at intermediate iterations (which are heuristics of potential practical significance). Finally we report the convergence and stability of obtained results. Fig. 18 depicts number of aware and rerouting drivers obtained throughout the 50 iterations. The number of aware drivers stabilizes already at fourth iteration, number of drivers making rerouting decision stabilizes around sixth iteration, while the relative gap of rerouting decisions logarithmically decreases to zero and falls below 10^{-3} in 10th iteration.

6. Summary and conclusions

The paper's objective was to investigate DTA capabilities of modelling consequences of unexpected events and to propose its modification allowing to represent rerouting phenomena. Based on DTA structure analysis we proposed Information Comply Model and integrated it into macroscopic DTA model of Gentile (2016). ICM originates from the thorough definition of

the drivers behaviour and their decision process. Dividing the decision process into two steps: becoming aware and deciding to reroute, allows handling two processes separately. Thanks to this multiple information sources available in contemporary traffic networks can be handled consistently. Impact of VMS, online sources, radio broadcasts can be handled jointly and further extended with drivers guessing about atypical situation from an atypical delay. Thanks to this, number of aware drivers (knowing about the event) can be consistently estimated as a function of location and time – which is crucial, as informing process can strongly depend on time (for the radio news) and location (for VMS). In our model only aware drivers may reroute, but not all aware drivers will reroute. Rerouting decision is driven by the rerouting utility, composed of two factors: avoiding losses and maximizing gains efficiently computed from the cost differences between typical and actual situations and further used in binomial logit to estimate rerouting probability in a given location and time. Thanks to this we can capture the rerouting only in cases when costs changed substantially and only at nodes with efficient alternatives. ICM is integrated in the modified demand model of DTA which now handles three drivers conditions of flow: unaware, aware and rerouted. Transitions between the conditions are handled with the awareness and rerouting models at each node of the propagation.

Final contribution of the paper, ICM-DTA, extends the DTA by calculating the network performance and traffic flows in cases of unexpected events. Proposed model overcomes the limitations of classic DTA methods (perfect information and predictive behaviour). It allows to efficiently compute travel times and vehicle flows for the unexpected event with acceptable computation burden. ICM can be computed in a standard DTA environment with input that is available and it provides a valuable, meaningful results in-line with what can be observed. In the case-study we demonstrated that the main objective was achieved (the rerouting phenomena was reproduced and computation times remained acceptable). In the parallel paper (Kucharski and Gentile, 2015) we show how ICM can be estimated with the field data to represent the actual phenomena, while in on-going research our aim is to use GPS *od* trajectories to understand the rerouting behaviour and calibrate rerouting model.

There are number of directions to pursue in context of this research. Starting from the awareness model, where more source types could be included and supported with empirical findings. Further research on how the data spreads online can bring a valuable insight, as well as testing if the awareness sources are correlated. Various types of information (raw fact, forecast, guidance, travel time, etc.) shall be included in the spreading process with various impact on the behaviour. The proposed model deals with a fixed *od* demand, while events affect whole activity plans which can change due to events, thus an extension to event-dependent *od* demand as defined in Dobler et al. (2012) might be a good improvement.

Also, the drivers' behaviour is probably more heterogenic and the single RCM for all users may be not valid, especially in unexpected cases. To this end one can introduce multiple demand classes, or use the perceived costs (derived from actual costs) for rerouting path. Similarly, one can substitute logit model with alternative discrete choice-model to overcome the issues of applying it sequentially along the trajectory, which in current version can lead to intractable results. The promising solutions seem to be: a) calculating rerouting utility up to the destination and determining the rerouting probability once, b) introducing a 'memory' into the sequential model by memorizing share of drivers that did not reroute and the utility that they rejected, c) discrete-choice models which can be applied sequentially (like recursive logit of Fosgerau et al. (2013)).

Finally, the biggest need for rerouting models is probably in real-time environment (TMC), yet to make ICM work in real-time, e.g. in rolling-horizon (Gentile et al., 2013) the awareness and rerouting models should be redefined for multiple independent events with independently computed awareness, rerouting utility and rerouting path.

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